PROVEIT RESEARCH CONTINUATION

Bilinear Alternating Tails and Harmonic Determinants

Rational certificates, Gamma identities, cutoff jets, and complete order-ten reconstruction

Prepared for Vladimir Reshetnikov

11 October 2026

Abstract. We develop exact identities from alternating Hurwitz tails and finite rational moment determinants. A classical Gamma quotient gives two bilinear endpoint laws with independent complex orders and shifts, all mixed generalized-Stieltjes derivatives, and anchored primitives expressed solely through Gamma ratios. We classify the rational two-adjacent-product certificates by an integer-orbit divisor condition and obtain explicit two-variable polygamma kernels. For the diagonal square law we calculate every nonpositive-order cutoff contact and every spectral jet, including the precise change under translating the cutoff. For real rational weights, we prove that total formal harmonic degree equals the generic coefficient-pencil rank, with only finitely many deficient numerical cutoffs. A positive finite-difference family realizes every rank at the same critical decay and has an exact hierarchy of spectral poles with Gamma-product leading coefficients. A critical two-particle cutoff has a depth-two polylogarithm generator, an explicit Euler primitive, and an ordinary-zeta integral endpoint. Finally, an exact polynomial congruence completes all smaller-order-ten harmonic-tail reconstructions and supplies an ordinary convergent polygamma identity family. A cyclotomic audit identifies a previously proved coefficient that should be removed from an open-test list. Proofs, exact computational certificates, independent diagnostics, source attribution, and further research targets are supplied.

Inspected repository revision

`da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`

Research status

New results are distinguished from inherited inputs and open questions. The order-ten argument includes an exact finite computer-assisted certificate. No arithmetic-independence or exhaustive priority claim is made.

Contents

1 Scope, principal results, and provenance **3**
1.1 Principal results . . . 3
1.2 What changes in the existing question list . . . 4
1.3 Classical input and the limits of the novelty claim . . . 4
1.4 Conventions and the rational harmonic decomposition . . . 5

2 A Gamma normalization for independently shifted Hurwitz tails **5**
2.1 The normalization and its difference equation . . . 6
2.2 Two exact bilinear endpoint laws . . . 6
2.3 All independent Stieltjes orders . . . 8
2.4 Anchored primitives and a pure Gamma square identity . . . 8
2.5 Scope and further targets . . . 9

3 Exactly when a rational adjacent-tail certificate exists **10**
3.1 The complete rational coefficient criterion . . . 10
3.2 Convergence and the finite endpoint . . . 11
3.3 Two independently shifted adjacent quadratic tails . . . 12
3.4 Every mixed integer order at independent shifts . . . 13

4 All cutoff jets of the alternating square law **13**
4.1 A finite identity valid at every spectral order . . . 14
4.2 The endpoint has only even inverse powers . . . 14
4.3 A convergent subtraction on every half-plane . . . 15
4.4 Every nonpositive-order contact and every spectral derivative . . . 16
4.5 The cutoff coordinate is part of the formula . . . 17
4.6 Explicit Gamma identities at order zero . . . 18

5 The harmonic degree is generically equal to the pencil rank **19**
5.1 An affine symmetric-pencil lemma . . . 19
5.2 Sharpness for moment matrices, including signed weights . . . 20
5.3 An exact test for exceptional cutoffs . . . 22
5.4 A positive example and a genuine signed exception . . . 22

6 A critical finite-difference hierarchy of every harmonic rank **23**
6.1 Finite moments and exact shifted-zeta closure . . . 23
6.2 Exact rank and the unique highest harmonic monomial . . . 24
6.3 Exact pole orders and the leading Laurent coefficients . . . 25
6.4 A Gamma product at the highest-order pole . . . 26
6.5 A complete two-particle formula and its Stieltjes finite parts . . . 27

7 A cutoff polylogarithm identity and its Euler primitive **28**
7.1 A convergent second Euler endpoint . . . 30

8 Complete reconstruction for smaller tail order ten **30**
8.1 Statement and exact meaning of reconstruction . . . 31
8.2 A finite five-dimensional definition of the numerator . . . 32
8.3 One additional bit removes the remaining finite obstruction . . . 34

8.4 A self-contained sufficient real-variable threshold 35
8.5 Completion, and an independent audit of the old exception 37
8.6 An ordinary convergent family of polygamma identities 37

9 A cyclotomic test case already determined by the manuscript 39

10 Further research questions with concrete mathematical targets 40
10.1 Reconstruction beyond order ten 40
10.2 Beyond the bilinear Gamma normalization 40
10.3 Harmonic pencils, spectral coefficients, and polylogarithms 41
10.4 A practical order of attack 41

11 Verification, source corrections, and integration 42
11.1 Proofs and exact computational certificates 42
11.2 Reproducible diagnostics 42
11.3 Corrections and status updates proposed for ProveIt 43
11.4 Integration boundary 43

1 Scope, principal results, and provenance

This article continues the polylogarithm and special-function programme in ProveIt at repository revision `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`, with particular attention to the unified manuscript and the five research packages then present in `docs/incoming`. The principal outcomes are exact identities and structural nonvanishing theorems. Bounds enter only as tools for one reconstruction proof.

The work has two related starting points. An alternating tail satisfies a first-order inhomogeneous recurrence; a Gamma quotient converts it to a homogeneous second-order recurrence with useful bilinear endpoint identities. A finite sum of rational moments has an affine harmonic matrix pencil; compressing onto a maximal-rank kernel determines its highest formal harmonic degree. In each case the finite object is established first. Infinite sums, derivatives, primitives, and Laurent coefficients follow with the relevant limiting operation stated.

1.1 Principal results

1. **Independent alternating tails.** For independently varying complex orders and shifts, a classical Gamma quotient produces both symmetric and alternating bilinear endpoint laws (Theorem 2.1). Every independent mixed spectral derivative at order one has an explicit formula in generalized Stieltjes constants (Theorem 2.3). Anchored integration produces a pure Gamma identity, including

$$\sum_{n=1}^{\infty} \frac{[h(n+2)-h(n+1)]^2}{h(n+1)h(n+2)} = 3 - \frac{\pi}{2} - \frac{4}{\pi}, \qquad h(z) = \sqrt{2}\,\frac{\Gamma((z+1)/2)}{\Gamma(z/2)}.$$

2. **A complete rational certificate classification.** Within the two-adjacent-product ansatz, the exact condition is $P(x)/Q(x) = cS(x)S(x-1)$ with rational S and outer color ± 1 (Theorem 3.1). Its divisor criterion gives obstructions as well as constructive certificates. A pair of adjacent quadratics yields explicit two-variable polygamma kernels.

3. **Every negative-order cutoff jet.** The square-law endpoint has an even inverse-power expansion. Its coefficient $c_m(s)$ determines every contact at $s = -m$, including all spectral derivatives, and gives an exact translation rule for a different cutoff coordinate (Theorem 4.4 and Corollary 4.5). For the natural cutoff, odd contacts vanish; at even m, the derivative threshold $r > m$ is sharp.

4. **Harmonic degree equals pencil rank for real rational weights.** The total formal degree is the generic rank over the rational cutoff field. Only finitely many numerical cutoffs can be deficient, and a precise kernel-compression test finds them (Theorem 5.2 and Corollary 5.3). A signed example shows why fixed-cutoff and generic claims differ.

5. **Every rank at the same critical decay.** A shorter finite-difference stencil realizes every harmonic rank $\rho = 1, \dots, r$ with decay $n^{-(2r-1)}$. Its spectral continuation has exact poles of orders $\rho, \rho-1, \dots, 1$ at $1, 2, \dots, \rho$, with explicit leading Laurent coefficients. The highest-order pole has a Gamma-product coefficient (Theorems 6.2 and 6.3). Its two-particle cutoff has a short depth-two polylogarithm generator, an explicit Euler primitive, and an ordinary-zeta integral endpoint (Section 7).

6. **Completion of all smaller-order-ten reconstructions.** For every even $q > 10$, the first q centered even moments of the $(10, q)$ harmonic-tail product have full formal rank q (Theorem 8.1). A new coefficientwise Bernoulli congruence closes the exact gap between two preceding reports. The resulting ordinary convergent polygamma identities are given in Corollary 8.4.

A separate audit establishes that the first complementary cyclotomic test coefficient already follows from two proved manuscript formulas (Proposition 9.1). This is a correction to its research status and cross-references, with credit retained by the original reductions.

1.2 What changes in the existing question list

Source question	Exact outcome here	Scope that remains
Alternating-tail report, Question 1	The order-ten exceptional progression is eliminated; the order-eight completion is inherited from the quartic report.	All smaller orders at most ten are now covered; larger smaller orders remain.
Same report, Question 4	Necessary and sufficient rational factorization for both adjacent products; independent quadratic kernels.	Certificates in larger classes and with more products.
Same report, Question 5	Entire endpoint continuation, normally convergent subtractions, all nonpositive-order contacts and their jets.	Multivariable contacts beyond the diagonal square law.
Same report, Question 6	Independently ordered and shifted *alternating* tails, with weights supplied by the Gamma normalization.	The question's ordinary Hurwitz product, arbitrary outer order, and its full moving-pole catalogue remain open.
Same report, Question 7	Its first displayed coefficient is already determined by manuscript identities; the implication is written out.	The complete complementary two-resolvent generator.
Finite-difference report, second research topic	Equality of degree and rank for every real rational weight over $\mathbb{R}(N)$, with a precise fixed-cutoff test.	Arbitrary complex weights and minimal depth under functional relations.
Finite-difference spectral family	All critical ranks, exact pole orders, and highest Laurent coefficients for shorter stencils.	General Gauss-ensemble overlapping resonances and unrestricted higher-rank pencil classification.

The source-question wording matters. In particular, a Gamma-weighted alternating identity is substantial progress on independent parameters, but it does not evaluate every independently weighted ordinary Hurwitz product. Likewise the rank theorem concerns formal harmonic coordinates, not arithmetic independence of the resulting numbers.

1.3 Classical input and the limits of the novelty claim

Alternating Hurwitz recurrences, their Stieltjes expansions, and the associated Gamma quotient are classical ingredients [8, 9]. The alternating-Hurwitz asymptotic expansions and their spectral derivatives are already treated by Hu–Kim [10]; the present contact theorem applies them to a specific quadratic endpoint and computes the resulting finite-part discrepancy. Products of zeta tails also have substantial prior literature, including Hoffman [11]. Rational difference equations and their rational solution algorithms long predate the specialized divisor calculation used here [12].

The determinant work starts from the finite-difference report's rational harmonic decomposition and its rank upper bound [3]. The order-ten proof reuses the fixed-size Bernoulli determinant and

the real-variable domination method with explicit credit [5, 2, 4]. Its new congruence, the complete coverage argument, and the resulting uniform reconstruction statement are separated from those reused ingredients.

Thus "new" in this article means a proved development absent from the inspected project corpus and not located in the targeted primary literature search. It is not a claim that an exhaustive priority search has excluded every previously published specialization. The proofs are the basis for the mathematical claims. Numerical agreement is used only to detect mistakes. No integer-relation fit is promoted to a theorem.

1.4 Conventions and the rational harmonic decomposition

The Hurwitz zeta function and generalized Stieltjes constants use

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{j\ge 0}\frac{(-1)^j\gamma_j(a)}{j!}u^j,\qquad \gamma_0(a)=-\psi(a).$$

All Gamma logarithms and complex powers in right half-planes use the holomorphic branches real on the positive axis. Shift derivatives and spectral derivatives are distinguished. An empty determinant or product is one. Ordinary harmonic numbers are $H_n^{(s)}=\sum_{k=1}^{n}k^{-s}$, with $H_0^{(s)}=0$; shifted cutoff sums are $H_N^{(q)}(a)=\sum_{n=0}^{N}(n+a)^{-q}$.

For clarity, here is the elementary decomposition behind the later degree theorem. Given a rational weight R without poles on the nonnegative integers, expand each $x^kR(x)$, $0\le k\le 2r-2$, into its polynomial part and principal parts. Polynomial sums are rational in N by Faulhaber's formula. Group the pole shifts modulo integer translation and choose one representative a_ν per orbit. A shifted principal-part sum differs from $H_N^{(q)}(a_\nu)$ by finitely many lower and upper endpoint terms, all rational in N. It follows exactly that

$$\left(\sum_{n=0}^{N}n^{i+j}R(n)\right)_{0\le i,j<r}=D_N+\sum_{\nu,q}H_N^{(q)}(a_\nu)C_{\nu,q},\tag{1.1}$$

where D_N is rational in N and the symmetric matrices $C_{\nu,q}$ do not depend on N. For a real rational weight, conjugate coordinates may be replaced by real and imaginary parts. Replacing these harmonic coordinates by independent indeterminates defines the formal pencil and its degree. This replacement is a bookkeeping construction; no algebraic-independence assertion is needed or made.

For infinite expressions, "finite part" always has a displayed meaning: either a Laurent coefficient in a spectral variable or the constant term in powers and logarithms of a specified cutoff. Different prescriptions are compared through their explicit endpoint contacts, rather than silently identified.

2 A Gamma normalization for independently shifted Hurwitz tails

The diagonal square law in the preceding report leaves open what happens when both tail orders and both shifts vary independently. A normalization by an elementary Gamma quotient gives a finite endpoint formula in all four variables. The resulting weight is determined by the two orders; an arbitrary additional outer power is not being evaluated here.

2.1 The normalization and its difference equation

For $\Re z > 0$ write

$$\mathcal{B}_s(z) = 2^{-s}\left\{\zeta\left(s, \frac{z}{2}\right) - \zeta\left(s, \frac{z+1}{2}\right)\right\}, \tag{2.1}$$

$$\ell(z) = \frac{1}{2}\log 2 + \log\Gamma\left(\frac{z+1}{2}\right) - \log\Gamma\left(\frac{z}{2}\right), \qquad h(z) = e^{\ell(z)}. \tag{2.2}$$

The difference in (2.1) has its removable value at $s = 1$ and is entire in s. All logarithms of Gamma in (2.2) are the holomorphic branches on the right half-plane that are real on the positive axis. Powers of h mean $e^{s\ell}$. The elementary recurrences give

$$\mathcal{B}_s(z) + \mathcal{B}_s(z+1) = z^{-s}, \qquad \ell(z) + \ell(z+1) = \log z, \qquad h(z)h(z+1) = z. \tag{2.3}$$

The alternating Hurwitz recurrence and its order derivatives are classical; see Hu–Kim [8]. The quotient h is also classical: in the notation of Wang–Hu–Kim [9], $h(z) = \sqrt{\pi/2}/\widetilde{\Gamma}(z)$.

Set

$$\tau_s(z) = h(z)^s\mathcal{B}_s(z), \qquad V_s(z) = \frac{z^s - (z-1)^s}{h(z)^{2s}} \quad (\Re z > 1). \tag{2.4}$$

Neither h nor this rewriting of $\mathcal{B}_s$ is asserted to be a new special function. Their useful feature is the homogeneous equation

$$\boxed{\tau_s(z-1) - \tau_s(z+1) = V_s(z)\tau_s(z).} \tag{2.5}$$

Indeed, multiplication of the two adjacent recurrences for $\mathcal{B}_s$ by the appropriate powers of h gives

$$\tau_s(z \pm 1) = h(z)^{-s} - \left(\frac{h(z \pm 1)}{h(z)}\right)^s \tau_s(z).$$

Subtracting and using (2.3) proves (2.5).

2.2 Two exact bilinear endpoint laws

Theorem 2.1 (Independent orders and shifts)**.** *Let $\Re a, \Re b > 0$. For $p, q \in \mathbb{C}$ put*

$$u_n = \tau_p(n+a+1), \qquad v_n = \tau_q(n+b+1), \qquad C_n = u_nv_{n+1} + u_{n+1}v_n, \quad D_n = u_nv_{n+1} - u_{n+1}v_n.$$

These definitions include $n = -1$. For every integer $N \geq 0$,

$$\sum_{n=0}^{N}\{V_p(n+a+1) + V_q(n+b+1)\}u_nv_n = C_{-1} - C_N, \tag{2.6}$$

$$\sum_{n=0}^{N}(-1)^n\{V_p(n+a+1) - V_q(n+b+1)\}u_nv_n = D_{-1} + (-1)^N D_N. \tag{2.7}$$

If $\Re(p+q) > 0$, both series converge absolutely and locally normally, and hence

$$\sum_{n\geq 0}\{V_p(n+a+1) + V_q(n+b+1)\}u_nv_n = \tau_p(a)\tau_q(b+1) + \tau_p(a+1)\tau_q(b), \tag{2.8}$$

$$\sum_{n\geq 0}(-1)^n\{V_p(n+a+1) - V_q(n+b+1)\}u_nv_n = \tau_p(a)\tau_q(b+1) - \tau_p(a+1)\tau_q(b). \tag{2.9}$$

All fixed mixed derivatives in p, q, a, b may be taken term by term in this domain. The second equality also holds as an ordinary, possibly conditional, series on $\Re(p+q) > -2$.

Proof. The homogeneous equation yields

$$C_{n-1} - C_n = \{V_p(n+a+1) + V_q(n+b+1)\}u_n v_n,$$

and

$$D_{n-1} + D_n = \{V_p(n+a+1) - V_q(n+b+1)\}u_n v_n.$$

The first identity telescopes. Multiplying the second by $(-1)^n$ also telescopes, since consecutive signs are opposite. This proves both finite identities, including their lower signs.

The standard Hurwitz expansion and the Gamma quotient expansion give, uniformly on compact sets of orders and shifts,

$$\begin{aligned} \mathcal{B}_s(z) &= \frac{1}{2} z^{-s} \left(1 + \frac{s}{2z} + O(z^{-2})\right), \\ h(z) &= z^{1/2} \left(1 - \frac{1}{4z} + O(z^{-2})\right), \\ \tau_s(z) &= \frac{1}{2} z^{-s/2} \left(1 + \frac{s}{4z} + O(z^{-2})\right), \\ V_s(z) &= \frac{s}{z} + O(z^{-2}). \end{aligned} \tag{2.10}$$

These expansions hold for arbitrary bounded complex s, with sufficiently many terms in the Hurwitz expansion when needed. Thus the summands in (2.8) and (2.9) are $O(n^{-1-\Re(p+q)/2})$, locally uniformly, and $C_N = O(N^{-\Re(p+q)/2})$. The leading terms cancel in D_N, giving $D_N = O(N^{-1-\Re(p+q)/2})$. This proves absolute normal convergence and the endpoint limits when $\Re(p+q) > 0$. Cauchy's formula gives the derivative assertion. Finally the finite identity (2.7) itself proves ordinary convergence on $\Re(p+q) > -2$, since its remaining endpoint tends to zero there. □

The Hurwitz and Gamma expansions invoked here are classical [14]; the contribution is the bilinear finite endpoint calculus built from them. On setting $p = q = s$ and $a = b$, division of (2.8) by two gives precisely

$$\sum_{n\geq 0} \{(n+a+1)^s - (n+a)^s\} \mathcal{B}_s(n+a+1)^2 = \mathcal{B}_s(a+1) - a^s \mathcal{B}_s(a+1)^2.$$

Thus the earlier square law is a diagonal specialization of this four-variable family.

Corollary 2.2 (The first boundary contact)**.** *For $p+q=0$, the ordinary symmetric series in (2.6) converges absolutely and satisfies*

$$\begin{aligned} &\sum_{n\geq 0} \{V_p(n+a+1) + V_{-p}(n+b+1)\} \tau_p(n+a+1)\tau_{-p}(n+b+1) \\ &= \tau_p(a)\tau_{-p}(b+1) + \tau_p(a+1)\tau_{-p}(b) - \frac{1}{2}. \end{aligned} \tag{2.11}$$

Proof. The n^{-1} terms in the sum of the two V's cancel, so the summand is $O(n^{-2})$. Equation (2.10) gives $C_N \to 1/2$. Now pass to the limit in (2.6). □

In particular the ordinary boundary value differs by $1/2$ from the analytic endpoint continuation from $\Re(p+q) > 0$. On the diagonal, the division by two explains the earlier $1/4$ contact. The present statement does not confuse those two notions of value.

2.3 All independent Stieltjes orders

Use the convention

$$\zeta(1+t,z)=\frac{1}{t}+\sum_{j\geq 0}\frac{(-1)^j\gamma_j(z)}{j!}t^j.$$

For $r\geq 0$ define the following explicit finite expressions:

$$T_r(z)=\frac{h(z)}{2}\sum_{j=0}^{r}\binom{r}{j}(\ell(z)-\log 2)^{r-j}(-1)^j\left\{\gamma_j\left(\frac{z}{2}\right)-\gamma_j\left(\frac{z+1}{2}\right)\right\}, \tag{2.12}$$

$$J_r(z)=\frac{1}{h(z)^2}\left\{z(\log z-2\ell(z))^r-(z-1)(\log(z-1)-2\ell(z))^r\right\}. \tag{2.13}$$

Here $\Re z>0$ suffices in (2.12), and $\Re z>1$ is used in (2.13). Direct coefficient extraction gives

$$T_r(z)=\partial_s^r\tau_s(z)|_{s=1}\,,\qquad J_r(z)=\partial_s^r V_s(z)|_{s=1}\,.$$

For example $T_0(z)=h(z)\mathcal{B}_1(z)$ and $J_0(z)=h(z)^{-2}$.

Theorem 2.3 (Every independent mixed Stieltjes jet)**.** *For $r,s\geq 0$ and $\Re a,\Re b>0$, write $z_n=n+a+1$ and $w_n=n+b+1$. Put*

$$X_{r,s}(z,w)=\sum_{j=0}^{r}\binom{r}{j}J_j(z)T_{r-j}(z)T_s(w),$$

$$Y_{r,s}(z,w)=\sum_{k=0}^{s}\binom{s}{k}T_r(z)J_k(w)T_{s-k}(w).$$

Then the following are ordinary absolutely convergent identities:

$$\sum_{n\geq 0}\{X_{r,s}(z_n,w_n)+Y_{r,s}(z_n,w_n)\}=T_r(a)T_s(b+1)+T_r(a+1)T_s(b), \tag{2.14}$$

$$\sum_{n\geq 0}(-1)^n\{X_{r,s}(z_n,w_n)-Y_{r,s}(z_n,w_n)\}=T_r(a)T_s(b+1)-T_r(a+1)T_s(b). \tag{2.15}$$

Proof. Apply $\partial_p^r\partial_q^s$ to (2.8) and (2.9) at $(p,q)=(1,1)$. The product rule gives the two finite binomial sums exactly. Each individual scalar summand in these sums is $O(n^{-2}\log^{r+s}(n+2))$, so no regularization or cancellation between divergent scalar series is implicit. □

For the first mixed jet, $r=s=1$, the symmetric summand is explicitly

$$\{J_0(z)+J_0(w)\}T_1(z)T_1(w)+J_1(z)T_0(z)T_1(w)+J_1(w)T_1(z)T_0(w).$$

Its sum is $T_1(a)T_1(b+1)+T_1(a+1)T_1(b)$. Thus independent first Stieltjes coefficients at different shifts already occur in an exact finite endpoint identity, and (2.14)– (2.15) give every higher pair without an inference from a single diagonal derivative.

2.4 Anchored primitives and a pure Gamma square identity

Differentiating (2.2) shows

$$\ell'(z)=\mathcal{B}_1(z),\qquad \tau_1(z)=h'(z). \tag{2.16}$$

Hence the $(p,q) = (1,1)$ laws have elementary primitives in the Gamma quotient itself. Let $A = [a_0, a_1]$ and $B = [b_0, b_1]$ be oriented real intervals with positive endpoints, and define

$$U_A(t) = h(t+a_1) - h(t+a_0), \qquad R_A(t) = \frac{1}{h(t+a_0)} - \frac{1}{h(t+a_1)},$$

with analogous notation for B. Double integration of (2.8) and (2.9), using $\int h'/h^2 = -1/h$, gives

$$\sum_{n\geq 1} \{R_A(n)U_B(n) + U_A(n)R_B(n)\} = U_A(0)U_B(1) + U_A(1)U_B(0), \tag{2.17}$$

$$\sum_{n\geq 1} (-1)^{n-1} \{R_A(n)U_B(n) - U_A(n)R_B(n)\} = U_A(0)U_B(1) - U_A(1)U_B(0). \tag{2.18}$$

These anchored identities have no unspecified integration constants. Uniform convergence on the compact rectangles justifies integration; alternatively $U_A(n) = O(n^{-1/2})$ and $R_A(n) = O(n^{-3/2})$ give absolute convergence directly.

Corollary 2.4 (A Gamma quotient increment square)**.** *For positive real a, b,*

$$\boxed{\sum_{n=1}^{\infty} \frac{\{h(n+b) - h(n+a)\}^2}{h(n+a)h(n+b)} = \{h(b) - h(a)\}\{h(b+1) - h(a+1)\}.} \tag{2.19}$$

Proof. Take the same interval $A = B = [a, b]$ in (2.17) and divide by two. There is also a particularly short finite certificate. If $U_n = h(n+b) - h(n+a)$, the recurrence $h(z)h(z+1) = z$ gives

$$U_{n-1}U_n - U_nU_{n+1} = \frac{U_n^2}{h(n+a)h(n+b)}.$$

Thus the sum through $n = N$ equals $U_0U_1 - U_NU_{N+1}$, and $U_NU_{N+1} = O(N^{-1})$ tends to zero. □

At $(a, b) = (1, 2)$ the right side is

$$3 - \frac{\pi}{2} - \frac{4}{\pi}. \tag{2.20}$$

Indeed $h(1) = \sqrt{2/\pi}$, $h(2) = \sqrt{\pi/2}$, and $h(3) = 2\sqrt{2/\pi}$. Equation (2.19) also extends holomorphically to $\Re a, \Re b > 0$, with the algebraic square as written and no complex conjugation.

2.5 Scope and further targets

The classical ingredients are the alternating Hurwitz recurrence, Gamma shift formula, and a two-solution discrete Lagrange identity. Products of zeta tails already have a broad literature, including Hoffman's multiple-zeta treatment [11]. The claims here are the displayed independently parameterized endpoint laws, their explicit mixed Stieltjes coefficients, and the anchored Gamma consequences, as additions to the inspected project material. A literature search did not locate these particular forms; this is not a proof of priority. No numerical independence assertion is used.

Two precise extensions remain. First, adjoining an independent outer power to (2.8) generally breaks the displayed telescoper; one should describe the additional coordinate and its moving polar hyperplanes. Second, at $p + q = -2m$ with $m > 0$, the exact upper endpoint provides a systematic continuation scheme, but the complete independent-order contact algebra is not established by Corollary 2.2 alone.

3 Exactly when a rational adjacent-tail certificate exists

The previous incoming report explicitly asks which pairs of polynomial denominators admit a rational certificate made from both adjacent tail products. The answer is a factorization criterion. It explains why generic shifted powers require a larger coefficient class, while also producing a simple two-variable polygamma kernel.

3.1 The complete rational coefficient criterion

Let $P, Q \in \mathbb{C}[x] \setminus \{0\}$ and write formally

$$E_{n-1} = E_n + \frac{(-1)^n}{P(n)}, \qquad F_{n-1} = F_n + \frac{(-1)^n}{Q(n)}.$$

The actual tails will later be $E_n = \sum_{k>n}(-1)^k/P(k)$ and $F_n = \sum_{k>n}(-1)^k/Q(k)$. For a fixed nonzero number t consider

$$K_n = t^n\{A(n)E_nF_{n+1} + B(n)E_{n+1}F_n\}, \qquad A, B \in \mathbb{C}(x). \tag{3.1}$$

By cancellation of linear-tail terms we mean cancellation of their formal coefficients after applying the two recurrences, with E_n, F_n treated as independent indeterminates. This is the precise certificate class being classified; possible identities exploiting additional relations between specific tails are not excluded.

Theorem 3.1 (Rational certificate classification)**.** *A nonzero rational coefficient pair (A, B) in (3.1) eliminates both linear-tail terms if and only if*

$$t \in \{1, -1\}, \qquad \frac{P(x)}{Q(x)} = cS(x)S(x-1) \tag{3.2}$$

for a nonzero rational function S and a nonzero constant c. For a fixed admissible t, all coefficient pairs are proportional to

$$A(x) = Q(x+1)S(x), \qquad B(x) = tQ(x+1)S(x+1). \tag{3.3}$$

If P/Q is constant, the $t = -1$ certificate gives the trivial zero identity for proportional actual tails.

Proof. Put $H = A + B$. Direct substitution into $K_n - K_{n-1}$ shows that linear-tail cancellation is equivalent to

$$\frac{A(x)}{Q(x+1)} = t^{-1}\frac{B(x-1)}{Q(x)}, \qquad \frac{B(x)}{P(x+1)} = t^{-1}\frac{A(x-1)}{P(x)}. \tag{3.4}$$

Neither coefficient can vanish unless both do. Eliminating B yields

$$\frac{A(x)}{A(x-2)} = t^{-2}\frac{Q(x+1)P(x)}{Q(x)P(x-1)}.$$

Both rational ratios apart from t^{-2} tend to one at infinity, so $t^2 = 1$. Put $R = P/Q$ and $S(x) = A(x)/Q(x+1)$. The remaining equation is

$$\frac{S(x)}{S(x-2)} = \frac{R(x)}{R(x-1)}.$$

Consequently $R(x)/(S(x)S(x-1))$ is a rational function of period one. A periodic rational function is constant: a pole would generate infinitely many poles, and a nonconstant polynomial cannot be periodic. This proves the necessity of (3.2). Conversely that factorization and (3.3) satisfy (3.4) by substitution. The quotient of two solutions for A is rational and of period two, so it is constant, proving uniqueness up to scalar. □

This elementary argument is a specialized rational difference-equation calculation, within the classical subject of rational summation and recurrence solutions [12]. It does not claim a new general algorithm for rational recurrences.

Corollary 3.2 (Root-orbit criterion)**.** *On each orbit $\alpha+\mathbb{Z}$ let m_j be the multiplicity of $x=\alpha+j$ as a zero of P/Q, with negative multiplicities for poles. A rational S in (3.2) exists if and only if*

$$\sum_{j\in\mathbb{Z}}(-1)^j m_j = 0 \quad \textit{on every such orbit.} \tag{3.5}$$

Proof. If s_j are the multiplicities for S, factorization is equivalent to $m_j = s_j + s_{j-1}$. Starting below the finite support of (m_j), the unique solution is $s_j = \sum_{k\le j}(-1)^{j-k}m_k$. It has finite support exactly under (3.5). The remaining nonzero multiplicative constant is absorbed into c. □

For example, for positive integer orders p, q and shifted powers $P(x)=(x+a)^p$, $Q(x)=(x+b)^q$, the criterion holds exactly when

$$p = q \quad \text{and} \quad a-b\in 2\mathbb{Z}. \tag{3.6}$$

If $a-b$ is not an integer the two roots occupy separate orbits. If it is an integer d, the condition is $p=(-1)^d q$, yielding (3.6). This is an obstruction only to (3.1), not to special-function endpoint evaluations in larger coefficient classes. The Gamma normalization in Section 2 supplies just such a larger class.

3.2 Convergence and the finite endpoint

Assume $\deg P, \deg Q \ge 1$, no positive integer zeros, and an admissible factorization. Put

$$H_t(x) = Q(x+1)\{S(x)+tS(x+1)\}, \qquad W_t(n) = t^n\{H_t(n)-tH_t(n-1)\}.$$

The certificate calculation gives exactly

$$K_n - K_{n-1} = W_t(n)E_nF_n. \tag{3.7}$$

Suppose also that the displayed coefficient values are finite on the summation lattice and at the lower endpoint; removable values may be taken. To state the endpoint without alternating signs in the tails, write

$$e_n = \sum_{j\ge 0}\frac{(-1)^j}{P(n+1+j)}, \qquad f_n = \sum_{j\ge 0}\frac{(-1)^j}{Q(n+1+j)}.$$

Then, whenever $S(0)$ is finite and nonzero,

$$\sum_{n\ge 0} W_t(n)e_nf_n = S(0)e_0 + \frac{t}{cS(0)}f_0 - tH_t(-1)e_0f_0. \tag{3.8}$$

The finite version subtracts

$$t^N\{A(N)e_N f_{N+1} + B(N)e_{N+1}f_N\}$$

from the right side of (3.8). This also gives a safe lower-cutoff version if one of the coefficient functions has a pole at one of the first lattice points.

For completeness, write $p = \deg P$, $q = \deg Q$. Factorization forces $d = (p+q)/2$ to be an integer at least one. Alternating summation gives $e_n = O(n^{-p})$, $f_n = O(n^{-q})$, while $A, B = O(n^d)$. For $t = 1$, the difference of H_1 is $O(n^{d-1})$. For $t = -1$, $S(n) - S(n+1)$ lowers the degree by at least one, so H_{-1} and its adjacent sum are $O(n^{d-1})$. Thus every series in (3.8) is absolutely convergent, and its upper endpoint is $O(N^{-d})$. For positive-order rational tails all endpoints reduce by partial fractions to finite polygamma values at half-shifts.

3.3 Two independently shifted adjacent quadratic tails

Define, for positive integer p,

$$D_p(z) = \mathcal{B}_p(z) - \mathcal{B}_p(z+1), \qquad D(z) = D_1(z) = 2\mathcal{B}_1(z) - \frac{1}{z}. \tag{3.9}$$

In particular

$$D(n+x) = \sum_{j\geq 0} \frac{(-1)^j}{(n+x+j)(n+x+j+1)}.$$

The denominators $P(k) = (k+x-1)(k+x)$ and $Q(k) = (k+y-1)(k+y)$ satisfy (3.2) with $S(k) = (k+x)/(k+y)$ and $c = 1$. This works for independent x, y; the roots within each quadratic are adjacent.

Theorem 3.3 (Finite two-variable polygamma kernels)**.** *For $\Re x, \Re y > 0$, both series below converge absolutely and locally normally:*

$$\sum_{n\geq 0}(2n+x+y)D(n+x)D(n+y) = \frac{1}{2}\left\{\frac{x}{y}D(x) + \frac{y}{x}D(y) - (2xy - x - y)D(x)D(y)\right\}, \tag{3.10}$$

$$\sum_{n\geq 0}(-1)^n D(n+x)D(n+y) = \frac{1}{2}D(x)D(y) + \frac{x^2D(x) - y^2D(y)}{2xy(x-y)}. \tag{3.11}$$

The quotient at $x = y$ is interpreted by continuity. In particular,

$$\sum_{n\geq 0}(-1)^n D(n+x)^2 = \frac{1}{2}D(x)^2 + \frac{D(x)}{x} + \frac{1}{2}D'(x). \tag{3.12}$$

Proof. For the stated P, Q, S, the two rational coefficients are

$$A(n) = (n+x)(n+y+1), \qquad B(n) = t(n+y)(n+x+1).$$

For $t = 1$, $H_1(n) - H_1(n-1) = 2(2n+x+y)$ and $H_1(-1) = 2xy - x - y$. Substitution in (3.8) and division by two proves (3.10). For $t = -1$, $H_{-1}(n) = x - y$ and $W_{-1}(n) = 2(x-y)(-1)^n$. The resulting endpoint is

$$\frac{x}{y}D(x) - \frac{y}{x}D(y) + (x-y)D(x)D(y),$$

which proves (3.11) for $x \neq y$. The locally normal series has a limit on the diagonal. Taking that limit on the right gives (3.12). The estimates $D(n+x) = O(n^{-2})$ prove all claimed convergence statements. □

For example, with $L = \log 2$,

$$\sum_{n\ge 0}(-1)^n D(n+1)^2 = 2L^2 - \frac{\pi^2}{12}, \tag{3.13}$$

$$\sum_{n\ge 0}(-1)^n D(n+1)D(n+2) = 1 - 2L^2, \tag{3.14}$$

$$\sum_{n\ge 0}(2n+3)D(n+1)D(n+2) = 2(1-L)^2. \tag{3.15}$$

These follow from $D(1) = 2L - 1$, $D(2) = 3/2 - 2L$, and $D'(1) = 1 - \pi^2/6$.

3.4 Every mixed integer order at independent shifts

Let $K(x, y)$ and $M(x, y)$ denote the right sides of (3.11) and (3.10), respectively. For $p, q \ge 1$ set

$$\mathfrak{D}_{p,q} = \frac{(-1)^{p+q-2}}{(p-1)!(q-1)!}\partial_x^{p-1}\partial_y^{q-1}.$$

The shift derivative formula gives $D^{(r)}(z) = (-1)^r r! D_{r+1}(z)$. Therefore

$$\boxed{\sum_{n\ge 0}(-1)^n D_p(n+x)D_q(n+y) = \mathfrak{D}_{p,q}K(x,y).} \tag{3.16}$$

The right side is a finite expression in polygamma values and a confluent divided difference; it contains no tail sum. Likewise

$$\sum_{n\ge 0}\big\{(2n+x+y)D_p(n+x)D_q(n+y) \\ - \mathbf{1}_{p\ge 2}D_{p-1}(n+x)D_q(n+y) - \mathbf{1}_{q\ge 2}D_p(n+x)D_{q-1}(n+y)\big\} = \mathfrak{D}_{p,q}M(x,y). \tag{3.17}$$

When an indicator is zero its corresponding term is absent. The formula follows by differentiating the linear weight as well as both tails, so no lower-order correction has been omitted.

The identities hold meromorphically away from the nonpositive-integer coordinate poles, with the diagonal $x = y$ removable. Their proof uses only rational cancellation and differentiation of normally convergent series. The obstruction (3.6) and the positive kernel family (3.16) concern different denominator classes and are fully compatible.

4 All cutoff jets of the alternating square law

The preceding independent-order identities grow from a finite endpoint identity. This section resolves Question 5 of the incoming alternating-tail report [2]: continue its ordinary square law through all nonpositive orders, calculate every cutoff contact, and distinguish the continued value from the constant term of the original finite sums. The alternating Hurwitz asymptotic expansion itself is classical; in particular, Hu and Kim give its remainder theory and spectral derivatives [10]. The contribution here is the explicit quadratic endpoint polynomial, its all-index contact formula, its sharp derivative threshold, and the resulting Gamma identities.

4.1 A finite identity valid at every spectral order

Throughout this section $\operatorname{Re} a > 0$. All powers and logarithms use the holomorphic branch in the right half-plane. Recall

$$\mathcal{B}_s(x) = 2^{-s}\left\{\zeta\left(s, \frac{x}{2}\right) - \zeta\left(s, \frac{x+1}{2}\right)\right\}. \tag{4.1}$$

It is entire in s, with the common Hurwitz pole at one cancelled. Put

$$F_s(x) = x^s \mathcal{B}_s(x)\mathcal{B}_s(x+1), \tag{4.2}$$

$$g_s(x) = \{x^s - (x-1)^s\}\mathcal{B}_s(x)^2, \tag{4.3}$$

$$S_N(s,a) = \sum_{n=0}^{N} g_s(n+a+1), \qquad X = N + a + 1. \tag{4.4}$$

Lemma 4.1 (Exact endpoint identity)**.** *For every complex s and every integer $N \geq 0$,*

$$S_N(s,a) = F_s(a) - F_s(X). \tag{4.5}$$

For $\operatorname{Re} s > 0$, the original series converges absolutely and locally normally to $F_s(a)$. Its unique entire continuation as a function of s is therefore $F_s(a)$.

Proof. The recurrence $\mathcal{B}_s(x) + \mathcal{B}_s(x+1) = x^{-s}$ gives

$$F_s(x-1) - F_s(x) = \{x^s - (x-1)^s\}\mathcal{B}_s(x)^2.$$

Telescoping proves the finite formula without any limiting argument. For $\operatorname{Re} s > 0$, the standard tail expansion gives $F_s(X) = O(X^{-\operatorname{Re} s})$ and $g_s(n+a+1) = O(n^{-\operatorname{Re} s - 1})$, uniformly on compact parameter sets. This proves the convergence assertion. Both sides of the finite identity are entire in s, so it already applies at every negative integer. □

An entire continuation is a function; a cutoff finite part is a summation prescription. Their agreement on a convergence half-plane does not make them identical at every exceptional parameter. We calculate the discrepancy instead of adopting an implicit convention.

4.2 The endpoint has only even inverse powers

Let $E_j(x)$ be the Euler polynomials, normalized by

$$\frac{2e^{xt}}{e^t + 1} = \sum_{j \geq 0} E_j(x)\frac{t^j}{j!}. \tag{4.6}$$

Thus $E_0(0) = 1$, $E_1(0) = -1/2$, and $E_{2j}(0) = 0$ for $j \geq 1$. For $j \geq 0$ define the rational polynomials

$$a_j(s) = \frac{(-1)^j E_j(0)}{2j!}(s)_j, \qquad (s)_0 = 1. \tag{4.7}$$

Define $c_0(s) = 1/4$ and, for $k \geq 1$,

$$c_k(s) = -\sum_{j=1}^{k-1} a_j(s)a_{k-j}(s). \tag{4.8}$$

In particular $c_{2j+1} = 0$. Every nonzero c_{2j} has degree $2j$. The first three nonconstant polynomials are

$$\begin{aligned} c_2(s) &= -\frac{s^2}{16}, \\ c_4(s) &= \frac{s^2(s+1)(s+2)}{96}, \\ c_6(s) &= -\frac{s^2(s+1)(s+2)(17s^2+99s+154)}{11520}. \end{aligned} \tag{4.9}$$

Proposition 4.2 (Even endpoint expansion)**.** *For every $K \geq 0$,*

$$F_s(x) = \sum_{j=0}^{K} c_{2j}(s) x^{-s-2j} + O(x^{-\operatorname{Re} s - 2K - 2}) \tag{4.10}$$

as $x \to +\infty$, locally uniformly in s and in bounded right-half-plane shifts of x. Every fixed spectral derivative has the differentiated expansion, with the corresponding finite powers of $\log x$. A derivative in x improves the remainder by one inverse power.

Proof. The alternating Hurwitz expansion is

$$\mathcal{B}_s(x) \sim x^{-s} \sum_{j \geq 0} a_j(s) x^{-j}.$$

One may obtain the needed uniform remainder by applying the ordinary Hurwitz Euler–Maclaurin expansion to the two terms in (4.1), using more terms than required and cancelling their common pole at one. The usual remainder integral is uniform on each compact s-set after enough terms have been retained [13, 10]. Its formal coefficients are also determined uniquely by $\mathcal{B}_s(x)+\mathcal{B}_s(x+1) = x^{-s}$: the formal operator $(1 + e^{\partial_x})^{-1}$ acting on x^{-s} gives (4.7). Cauchy's formula on a slightly larger compact set justifies every fixed spectral derivative of the remainder. The derivative identity $\partial_x \mathcal{B}_s(x) = -s\mathcal{B}_{s+1}(x)$ supplies the corresponding derivative remainder in x.

Write $A = x^s \mathcal{B}_s(x)$. By the same recurrence,

$$F_s(x) = x^{-s}(A - A^2) = x^{-s} \left\{ \frac{1}{4} - \left(A - \frac{1}{2} \right)^2 \right\}.$$

All positive even coefficients of A vanish. Thus $A - 1/2$ has only odd inverse powers, its square has only even inverse powers, and (4.8) gives every coefficient. Multiplying finite expansions with their uniform remainders proves the assertion, rather than assuming convergence of the infinite asymptotic series. □

4.3 A convergent subtraction on every half-plane

For $K \geq 0$ set

$$P_K(s, x) = \sum_{j=0}^{K} c_{2j}(s) x^{-s-2j}. \tag{4.11}$$

Theorem 4.3 (Normally convergent continuation)**.** *On $\operatorname{Re} s > -2K - 2$, $\operatorname{Re} a > 0$, the following series converges absolutely and locally normally, including after every fixed parameter derivative:*

$$\begin{aligned} \sum_{n=0}^{\infty} \{ g_s(n + a + 1) - P_K(s, n + a) + P_K(s, n + a + 1) \} \\ = F_s(a) - P_K(s, a). \end{aligned} \tag{4.12}$$

Consequently adding $P_K(s,a)$ gives the same entire continuation $F_s(a)$ on all these overlapping half-planes.

Proof. Subtract the finite telescoping difference of P_K from (4.5). The remaining upper endpoint is $F_s(X) - P_K(s,X) = O(X^{-\operatorname{Re} s-2K-2})$ and tends to zero on the stated half-plane. Each summand is a unit difference of this remainder; the x-derivative version of Proposition 4.2 bounds it by $O(n^{-\operatorname{Re} s-2K-3})$. Uniformity on compact sets and Cauchy's formula give local normal convergence and all fixed parameter derivatives. □

This is an ordinary convergent-series realization of the continued function. It explicitly specifies what was subtracted. It neither assigns an ordinary value to a divergent unmodified series nor discards the exact upper endpoint.

4.4 Every nonpositive-order contact and every spectral derivative

For a fixed nonnegative integer m and derivative order r, expand $\partial_s^r S_N(s,a)|_{s=-m}$ in powers of $X = N + a + 1$ and $\log X$. Its *natural-cutoff finite part* is the coefficient of $X^0(\log X)^0$. Denote it by FP_X. The preceding expansion supplies such an asymptotic expansion to any fixed order, with a remainder tending to zero after enough terms.

Theorem 4.4 (Complete contact formula)**.** *For every $m, r \geq 0$ and $\operatorname{Re} a > 0$,*

$$\boxed{\mathrm{FP}_X\ \partial_s^r S_N(s,a)|_{s=-m} = \partial_s^r F_s(a)|_{s=-m} - c_m^{(r)}(-m).} \tag{4.13}$$

The contacts have the following exact properties:

1. *If m is odd, every natural-cutoff contact is zero.*
2. *If m is even, the contact vanishes for $r > m$, and the threshold is sharp:*

$$c_m^{(m)}(-m) = -\frac{1}{2}E_{m+1}(0) \neq 0. \tag{4.14}$$

3. *At derivative order zero,*

$$c_m(-m) = \frac{1}{4}[x^m]\{E_m(x)E_m(x+1)\}. \tag{4.15}$$

For even $m = 2j \geq 2$ this becomes

$$c_{2j}(-2j) = -\frac{1}{4}\sum_{\substack{1\leq k<2j\\ k\ odd}} \binom{2j}{k}^2 E_k(0)E_{2j-k}(0), \tag{4.16}$$

which is nonzero and has sign $(-1)^j$.

Proof. Differentiate the endpoint expansion before setting $s = -m$. For each k,

$$\partial_s^r\{c_k(s)X^{-s-k}\}\Big|_{s=-m} = X^{m-k}\sum_{\ell=0}^{r}\binom{r}{\ell}c_k^{(\ell)}(-m)(-\log X)^{r-\ell}.$$

Only $k = m$ has exponent zero, and only $\ell = r$ has logarithmic degree zero. The finite identity therefore gives (4.13). If m is odd, c_m is identically zero. Degree at most m gives the vanishing for $r > m$.

For the exact top coefficient, replace each $(s)_j$ in (4.7) by its leading monomial s^j. The resulting generating series is

$$A(t) = \frac{1}{1+e^{-t}} = \frac{1}{2} + \frac{1}{2}\tanh(t/2).$$

The endpoint's leading-coefficient series is

$$A(t) - A(t)^2 = A'(t) = \frac{1}{4\cosh^2(t/2)}.$$

Its coefficient of t^m is $(-1)^{m+1}E_{m+1}(0)/(2m!)$; for even m this proves (4.14). Odd Euler values at zero are nonzero, for example by their elementary Bernoulli-number formula.

Finally $\mathcal{B}_{-m}(x) = E_m(x)/2$ is the classical Euler specialization [10]. Hence $F_{-m}(x) = x^{-m}E_m(x)E_m(x+1)/4$. Its constant coefficient gives (4.15). For even $m \geq 2$, use $E_m(x+1) = 2x^m - E_m(x)$ and $E_m(0) = 0$ to obtain (4.16). Each summand in that convolution has sign $(-1)^{j+1}$, so its negative quarter has the asserted nonzero sign. □

m	0	2	4	6	8
$c_m(-m)$	1/4	−1/4	1	−43/4	230
$c_m^{(m)}(-m)$	1/4	−1/8	1/4	−17/16	31/4

At $s = 0$ every unmodified summand is exactly zero, whereas $F_0(a) = 1/4$. Formula (4.13) gives $0 = 1/4 - 1/4$. At $s = -2$, the summand simplifies to $-(2n + 2a + 1)/4$, so

$$S_N(-2, a) = \frac{a^2 - X^2}{4}, \qquad F_{-2}(a) = \frac{a^2 - 1}{4}.$$

The natural-cutoff constant is $a^2/4$, exactly the continued value minus the contact $-1/4$. The first two derivative contacts at this same order are $c_2'(-2) = 1/4$ and $c_2''(-2) = -1/8$.

4.5 The cutoff coordinate is part of the formula

The usual integer cutoff N and the natural cutoff $X = N + a + 1$ have different constant terms. The conversion is finite and explicit. Let $T = X - h$ for a fixed complex number h. Define the degree-at-most-m polynomial in s

$$C_m(s; h) = \sum_{k=0}^{m} c_k(s) \binom{-s-k}{m-k} h^{m-k}. \tag{4.17}$$

The generalized binomial here is its polynomial expression, so its spectral derivatives at integers are unambiguous.

Corollary 4.5 (Translated-cutoff contacts)**.** *For all $m, r \geq 0$,*

$$\mathrm{FP}_T\, \partial_s^r S_N(s, a)|_{s=-m} = \partial_s^r F_s(a)|_{s=-m} - \partial_s^r C_m(s; h)|_{s=-m}\,. \tag{4.18}$$

Take $h = a + 1$ to obtain the finite part in powers of N. Every translated contact still vanishes for $r > m$. At $r = 0$ its value is $\sum_{k=0}^{m} c_k(-m)h^{m-k}$.

Proof. Expand $X^{-s-k} = T^{-s-k}(1 + h/T)^{-s-k}$ by its finite asymptotic binomial expansion. At $s = -m$, the unique term capable of contributing a constant has binomial index $m-k$. Differentiating first retains its spectral coefficient and logarithms; the logarithm-free constant is exactly the derivative in (4.18). Terms with $k > m$ already decay. The polynomial degree bound follows from $\deg c_k \leq k$ and $\deg_s \binom{-s-k}{m-k} = m - k$. □

For example the natural contact at $s = -1$ is zero but the contact in $T = X - h$ is $h/4$. Thus "odd negative orders have no contact" is correct only with the displayed natural coordinate. Translation does not preserve that parity assertion.

One can also write the highest derivative in closed form. Define Euler polynomials of order two by

$$\frac{4e^{xt}}{(e^t+1)^2} = \sum_{m\ge 0} E_m^{(2)}(x)\frac{t^m}{m!}.$$

The same leading-coefficient calculation gives

$$\partial_s^m C_m(s;h) = \frac{(-1)^m}{4} E_m^{(2)}(h+1). \tag{4.19}$$

This includes (4.14) when $h = 0$ and m is even.

4.6 Explicit Gamma identities at order zero

The classical Lerch formula gives

$$\partial_s \mathcal{B}_s(x)|_{s=0} = -\log h(x), \qquad h(x) = \sqrt{2}\,\frac{\Gamma((x+1)/2)}{\Gamma(x/2)}. \tag{4.20}$$

Put

$$\rho(x) = \frac{1}{2}\log x - \log h(x) = \log\left\{\sqrt{\frac{x}{2}}\,\frac{\Gamma(x/2)}{\Gamma((x+1)/2)}\right\}. \tag{4.21}$$

The normalized endpoint immediately gives

$$\begin{aligned} \partial_s F_s(a)\big|_{s=0} &= -\frac{1}{4}\log a, \\ \partial_s^2 F_s(a)\Big|_{s=0} &= \frac{1}{4}\log^2 a - 2\rho(a)^2. \end{aligned} \tag{4.22}$$

Indeed, with $D_s(a) = a^s\mathcal{B}_s(a) - 1/2$, $F_s(a) = a^{-s}(1/4 - D_s(a)^2)$, $D_0 = 0$, and $D_0' = \rho(a)$. In particular the natural finite part of the second spectral derivative at $a = 1, s = 0$ is $-\frac{1}{2}\log^2(\pi/2)$.

There is an ordinary convergent version with a useful polarization. For $\operatorname{Re} a, \operatorname{Re} b > 0$ write

$$\delta_a(n) = \log\frac{n+a+1}{n+a}, \qquad \rho_a(n) = \rho(n+a).$$

Corollary 4.6 (A bilinear logarithmic Gamma sum)**.** *For* $\operatorname{Re} a, \operatorname{Re} b > 0$*,*

$$\sum_{n=0}^{\infty}\{\delta_a(n)\rho_b(n) + \delta_b(n)\rho_a(n) - \tfrac{1}{2}\delta_a(n)\delta_b(n)\} = 2\rho(a)\rho(b). \tag{4.23}$$

The convergence is absolute and locally normal. At $a = b$ *this reduces to*

$$\sum_{n=0}^{\infty}\delta_a(n)\left\{\rho_a(n) - \frac{1}{4}\delta_a(n)\right\} = \rho(a)^2. \tag{4.24}$$

In particular the value at $a = 1$ *is* $\frac{1}{4}\log^2(\pi/2)$*.*

Proof. The recurrence $h(x)h(x+1)=x$ implies $\rho(x)+\rho(x+1)=\frac{1}{2}\log((x+1)/x)$. Multiply the two shifted recurrences and rearrange:

$$2\{\rho_a(n)\rho_b(n)-\rho_a(n+1)\rho_b(n+1)\}=\delta_a(n)\rho_b(n)+\delta_b(n)\rho_a(n)-\tfrac{1}{2}\delta_a(n)\delta_b(n).$$

Telescoping proves the formula. Stirling's expansion gives $\rho(n+a)=O(n^{-1})$ with a unit difference $O(n^{-2})$, so the summand is $O(n^{-3})$ locally uniformly and the upper endpoint vanishes. The square case follows by division by two. Equivalently it is the second derivative of the normally subtracted identity (4.12) at $s=0$. □

Every fixed shift derivative of (4.23) is an ordinary convergent identity with rational functions, logarithms and polygamma functions. For higher spectral derivatives at zero or at $-m$, the finite formula (4.13) uses only finitely many Hurwitz order derivatives at the same integer. No independent arithmetic reduction of those order derivatives is asserted.

5 The harmonic degree is generically equal to the pencil rank

The finite-difference determinant report proves an upper bound for the harmonic degree and asks when that bound is strict. For real rational weights, the answer is particularly clean: it is never strict over the field of rational functions in the cutoff. At a specified integer cutoff it can be strict, but only at finitely many cutoffs. Positivity removes these exceptions once enough nodes have been included.

5.1 An affine symmetric-pencil lemma

Let $C_1,\dots,C_t$ be real symmetric $r\times r$ matrices, write

$$C(X)=\sum_{j=1}^{t}X_jC_j,\qquad \rho=\operatorname{rank}_{\mathbb{R}(X)}C(X),\qquad q=r-\rho,$$

and let D be another real symmetric matrix. The variables X_j are formal independent variables. Their introduction makes no assertion about algebraic independence of special functions.

Lemma 5.1 (Compression and the highest degree)**.** *The polynomial $P(X)=\det(D+C(X))$ has total degree at most ρ. Choose any $v\in\mathbb{R}^t$ such that $C(v)$ has rank ρ, and let K be a real $r\times q$ matrix whose columns form its kernel. Then*

$$K^{\mathsf{T}}C_jK=0\qquad(1\le j\le t).\tag{5.1}$$

If $K^{\mathsf{T}}DK$ is nonsingular, P has total degree exactly ρ. For a generic maximal-rank direction, nonsingularity of this compression is also necessary.

Proof. In each determinant term, selecting more than ρ columns from $C(X)$ gives zero, which proves the upper bound. Put $A=C(v)$ and choose an invertible real matrix $S=[U\ K]$ so that

$$S^{\mathsf{T}}AS=\begin{pmatrix}A_0&0\\0&0\end{pmatrix},\qquad\det A_0\neq0.$$

For every j, the rank of $A+zC_j$ is at most ρ for every real z: every $(\rho+1)$-minor of the whole pencil is identically zero. For sufficiently small z, its upper-left $\rho\times\rho$ block remains invertible. The Schur

complement of this block must therefore vanish identically. Its coefficient of z is exactly $K^{\mathsf{T}}C_jK$, proving (5.1).

The coefficient of z^ρ in $\det(D+zA)$ is

$$\frac{\det A_0}{(\det S)^2}\det(K^{\mathsf{T}}DK). \tag{5.2}$$

This follows by taking the zA_0 entry in each of the first ρ rows and columns after the congruence by S; the remaining block is $K^{\mathsf{T}}DK$. Thus a nonzero compression proves degree ρ. Conversely, if the degree-ρ homogeneous part is nonzero, it does not vanish at a generic real maximal-rank direction, and (5.2) makes that compression nonsingular. The conventions for $q=0$ are that K has no columns and its compressed determinant is one. □

There is a direct symbolic version that avoids expanding the full $r\times r$ determinant. Choose a nonsingular principal $\rho\times\rho$ submatrix of $C(X)$ and, after a simultaneous permutation of rows and columns, write

$$C(X)=\begin{pmatrix}A & B\\ B^{\mathsf{T}} & F\end{pmatrix}.$$

Over $\mathbb{R}(X)$, the rank condition gives $F=B^{\mathsf{T}}A^{-1}B$. Set

$$K(X)=\begin{pmatrix}-A^{-1}B\\ I_q\end{pmatrix}. \tag{5.3}$$

Then the highest homogeneous part has the exact expression

$$[P]_\rho(X)=\det A\,\det\big(K(X)^{\mathsf{T}}DK(X)\big). \tag{5.4}$$

Although the displayed factors are rational functions, their product is the indicated homogeneous polynomial. The expression remains valid when it vanishes. A symmetric matrix of rank ρ over a field of characteristic zero has a nonsingular principal minor of size ρ; this follows, for example, by symmetric elimination with 1×1 and 2×2 nonsingular pivots.

5.2 Sharpness for moment matrices, including signed weights

Let $w_n\in\mathbb{R}$ be a sequence and put

$$M_N=\sum_{n=0}^{N}w_n v(n)v(n)^{\mathsf{T}},\qquad v(n)=(1,n,\dots,n^{r-1})^{\mathsf{T}}. \tag{5.5}$$

Suppose that a decomposition

$$M_N=D_N+\sum_{j=1}^{t}h_j(N)C_j \tag{5.6}$$

holds with real $h_j(N)$ and real symmetric matrices D_N,C_j, where the C_j are independent of N. Define $P_N(X)=\det(D_N+C(X))$ and let δ_N be its total degree, with degree $-\infty$ assigned to the zero polynomial.

Theorem 5.2 (Only finitely many deficient cutoffs)**.** *Assume there are an integer $N_*\ge0$ and $\epsilon\in\{1,-1\}$ such that $\epsilon w_n>0$ for every integer $n>N_*$. Then*

$$\delta_N\le\rho\quad\textit{for all } N,$$

and at most $r(r-\rho)$ integers $N \geq N_*$ satisfy $\delta_N < \rho$. If all nonzero w_n have the same sign and at least r distinct nodes of positive absolute weight have been included by cutoff N, then $\delta_N = \rho$ at that cutoff.

If in addition D_N has entries rational in N, then

$$\deg_X \det(D_N + C(X)) = \rho \qquad \text{over } \mathbb{R}(N). \tag{5.7}$$

Proof. Choose one real maximal-rank direction and its kernel matrix K as in Lemma 5.1. Such a real direction exists because a nonzero maximal minor is a nonzero real polynomial. Equation (5.1) and (5.6) imply

$$G_N := K^{\mathsf{T}} D_N K = K^{\mathsf{T}} M_N K = \sum_{n=0}^{N} w_n p(n) p(n)^{\mathsf{T}}, \qquad p(n) = K^{\mathsf{T}} v(n). \tag{5.8}$$

The components of p are linearly independent real polynomials of degree at most $r-1$.

For $N \geq N_*$, the matrix

$$\epsilon(G_{N+r} - G_N) = \sum_{n=N+1}^{N+r} \epsilon w_n p(n) p(n)^{\mathsf{T}}$$

is positive definite. Indeed, a nonzero linear combination of the components of p is a nonzero polynomial of degree at most $r-1$; it cannot vanish at all r displayed nodes. Along any one of the r residue classes of cutoffs modulo r, the ordered eigenvalues of ϵG_N therefore increase strictly. Each of the $q = r - \rho$ eigenvalues can equal zero at most once. Consequently at most q cutoffs in each residue class have singular G_N, and at most rq cutoffs in total do. By Lemma 5.1, degree deficiency can occur only at those cutoffs. This also treats $q = 0$, for which there are no exceptions.

If the weights in the cutoff have one sign and at least r distinct nonzero nodes, the same polynomial argument makes G_N definite directly. Finally, in the rational case $\det G_N$ is a rational function of N. It is nonzero at some integer cutoff by the preceding argument, hence is not the zero rational function. Formula (5.2) proves (5.7). □

Corollary 5.3 (Resolution for real rational weights)**.** *Let R be a nonzero real rational function having no pole at a nonnegative integer. Form its finite moment matrices and their harmonic decomposition as in the finite-difference determinant report. Then their formal harmonic degree over $\mathbb{R}(N)$ is exactly the generic harmonic-pencil rank. At most finitely many nonnegative integer cutoffs have smaller degree. For $R(n) > 0$ at all nonnegative integers, every $N \geq r - 1$ has the exact generic degree.*

Proof. A nonzero real rational function has constant nonzero sign for all sufficiently large positive arguments. The residue construction gives the decomposition (5.6) with rational D_N. Apply Theorem 5.2. When poles and residues are represented using nonreal conjugate pairs, replace each conjugate pair of formal harmonic coordinates by its formal real and imaginary parts. This invertible linear change makes the coefficient matrices real and preserves both generic rank and total degree. □

The real hypothesis in this proof is substantive: it supplies definiteness of the sufficiently late increments. No corresponding theorem for an arbitrary complex rational weight is asserted here. Likewise, formal degree does not by itself prove a minimal multiple-polylogarithm depth under functional relations.

5.3 An exact test for exceptional cutoffs

In the generic kernel formula (5.3), replace K by the polynomial matrix

$$\widetilde{K}(X) = \begin{pmatrix} -\operatorname{adj}(A)B \\ (\det A)I_q \end{pmatrix}.$$

After clearing denominators in N, compute the polynomial

$$Q(N, X) = \det\big(\widetilde{K}(X)^{\mathsf{T}} D_N \widetilde{K}(X)\big). \tag{5.9}$$

For an admissible cutoff N_0, degree is deficient if and only if $Q(N_0, X)$ is identically zero. Equivalently, all its coefficients in X vanish. Over an exact coefficient field, their greatest common divisor as polynomials in N gives precisely the exceptional cutoffs among its admissible nonnegative integer roots. Denominator zeros must be excluded or checked from the original finite moment formula. This uses a determinant of size $r - \rho$, not an expansion of the original determinant. A single fixed maximal-rank direction gives a cheaper finite candidate list, which the generic test can then filter.

5.4 A positive example and a genuine signed exception

Take $r = 3$ and first let

$$R(x) = \frac{1}{x+1} + \frac{1}{x+2}.$$

There is one harmonic variable $X = H_N^{(1)}(1)$, and its coefficient is

$$C = \begin{pmatrix} 2 & -3 & 5 \\ -3 & 5 & -9 \\ 5 & -9 & 17 \end{pmatrix}, \qquad \operatorname{rank} C = 2, \qquad K = (2, 3, 1)^{\mathsf{T}}.$$

Since $K^{\mathsf{T}} v(n) = (n+1)(n+2)$, the leading coefficient is

$$[P_N]_2 = \frac{(N+1)(N+2)^2(N+3)}{2} X^2. \tag{5.10}$$

Here the top-left 2×2 minor of C has determinant one.

For the signed weight

$$R(x) = \frac{1}{x+1} - \frac{7}{5(x+2)} = \frac{3 - 2x}{5(x+1)(x+2)}, \tag{5.11}$$

the same kernel has compressed moment

$$K^{\mathsf{T}} D_N K = -\frac{(N-2)(N+1)(N+2)(N+3)}{10}.$$

The relevant principal minor of C has determinant $-7/5$, so

$$[P_N]_2 = \frac{7(N-2)(N+1)(N+2)(N+3)}{50} X^2. \tag{5.12}$$

The only exceptional nonnegative cutoff is $N = 2$, where the entire formal polynomial is

$$P_2(X) = \frac{620X - 1137}{500}. \tag{5.13}$$

Substituting the actual $X = H_2^{(1)}(1) = 11/6$ gives $-1/1500$, also equal to the three-node Vandermonde square times $R(0)R(1)R(2)$. Thus a fixed-cutoff exception is real, even when the finite moment determinant itself is nonzero. It does not contradict generic sharpness over $\mathbb{R}(N)$.

6 A critical finite-difference hierarchy of every harmonic rank

The original finite-difference construction uses a stencil of order $2r-2$ and confines its special-function dependence to one moment entry. Shorter stencils give an explicit hierarchy whose harmonic ranks run through every value from one to r, while the integer-order weights retain the same critical decay. Its spectral continuation has a completely determined staircase of pole orders. This is a spectral finite-difference family; the more general overlapping resonances of the earlier Gauss ensemble are a separate question.

Fix integers

$$r \geq 2, \qquad 1 \leq \rho \leq r, \qquad d = 2r-2, \qquad h = d-\rho+1 = 2r-\rho-1. \tag{6.1}$$

For $\Re a > 0$, use the right-half-plane logarithm to define

$$w_n^{[h]}(s,a) = \frac{1}{h!}\sum_{j=0}^{h}(-1)^j\binom{h}{j}(n+a+j)^{-s}. \tag{6.2}$$

The corresponding ordered ensemble is

$$Z_{r,\rho}(s,a) = \sum_{0\leq n_1<\cdots<n_r} \Delta(n_1,\ldots,n_r)^2 \prod_{i=1}^{r} w_{n_i}^{[h]}(s,a). \tag{6.3}$$

There is no additional $1/r!$ because the indices are strictly ordered.

6.1 Finite moments and exact shifted-zeta closure

Define finite polynomials and endpoint functions by

$$b_k(a;h) = \frac{1}{h!}\sum_{j=0}^{h}(-1)^j\binom{h}{j}(-a-j)^k, \tag{6.4}$$

$$E_k(s,a;h) = -\frac{1}{h!}\sum_{j=1}^{h}(-1)^j\binom{h}{j}\sum_{\ell=0}^{j-1}(\ell-j)^k(\ell+a)^{-s}. \tag{6.5}$$

The convention $0^0 = 1$ applies to zeroth moments. Polynomial annihilation gives

$$b_0 = \cdots = b_{h-1} = 0, \qquad b_h = 1. \tag{6.6}$$

For $0 \leq \ell \leq \rho-1$ let $C_\ell(a;h)$ be the symmetric Hankel matrix with entries

$$(C_\ell)_{ij} = \binom{i+j}{\ell} b_{i+j-\ell}(a;h), \qquad 0 \leq i,j < r, \tag{6.7}$$

with a term set to zero if its lower index is negative or if $\ell > i+j$. Also let $E(s,a;h) = (E_{i+j}(s,a;h))_{i,j<r}$.

Theorem 6.1 (A finite shifted-zeta determinant)**.** *The ensemble* (6.3) *is absolutely and locally normally convergent for $\Re s > \rho$, $\Re a > 0$, and there*

$$\boxed{Z_{r,\rho}(s,a) = \det\left(E(s,a;h) + \sum_{\ell=0}^{\rho-1}\zeta(s-\ell,a)C_\ell(a;h)\right).} \tag{6.8}$$

The right side gives its meromorphic continuation to all $s \in \mathbb{C}$. Its only possible poles are $1,2,\ldots,\rho$.

Proof. For $\Re s > 0$ the weight has the integral representation

$$w_n^{[h]}(s,a) = \frac{1}{h!\Gamma(s)} \int_0^\infty t^{s-1} e^{-(n+a)t} (1-e^{-t})^h \, dt. \tag{6.9}$$

Repeated finite differences, or expansion of this kernel at zero, give locally uniformly

$$w_n^{[h]}(s,a) = \frac{(s)_h}{h!}(n+a)^{-s-h} + O((n+1)^{-\Re s-h-1}). \tag{6.10}$$

For ordered indices, $|\Delta(\boldsymbol{n})|^2 \le \prod_i (n_i+1)^{2(i-1)}$. The least decaying factor is summable when $\Re s + h - d > 1$, precisely $\Re s > \rho$.

For a moment of degree $k \le d$, initially take $\Re s > d+1$ so that all separately shifted sums converge. Shift the index by j and expand

$$(n-j)^k = \sum_{\ell=0}^{k} \binom{k}{\ell} (-a-j)^{k-\ell} (n+a)^\ell.$$

The lower endpoints give (6.5), and the infinite shifted sums give

$$\sum_{n\ge 0} n^k w_n^{[h]}(s,a) = E_k(s,a;h) + \sum_{\ell=0}^{k-h} \binom{k}{\ell} b_{k-\ell}(a;h)\zeta(s-\ell,a). \tag{6.11}$$

An empty sum is zero. Both sides are holomorphic on their common normal-convergence half-plane, so the identity theorem extends this identity to $\Re s > \rho$. Finite Cauchy–Binet and passage to the normal limit give (6.8). Each endpoint function is entire in s, and every Hurwitz factor has just its displayed pole. □

For later cutoff use, define $H_N^{(z)}(a) = \sum_{n=0}^{N} (n+a)^{-z}$ for every complex z and

$$T_{k,N}(s,a;h) = \frac{1}{h!} \sum_{j=1}^{h} (-1)^j \binom{h}{j} \sum_{\ell=N+1}^{N+j} (\ell-j)^k (\ell+a)^{-s}. \tag{6.12}$$

The exact finite counterpart of (6.11) is

$$\sum_{n=0}^{N} n^k w_n^{[h]}(s,a) = E_k + T_{k,N} + \sum_{\ell=0}^{k-h} \binom{k}{\ell} b_{k-\ell} H_N^{(s-\ell)}(a). \tag{6.13}$$

It holds for every s, including orders at which the infinite ensemble has a pole. It follows from the finite index-shift identity; no divergent series is split in its proof.

6.2 Exact rank and the unique highest harmonic monomial

Theorem 6.2 (Every rank at the same critical decay)**.** *At $s = \rho$ and real $a > 0$, the rational weight $R_{r,\rho,a}(n) = w_n^{[h]}(\rho,a)$ is positive and obeys*

$$R_{r,\rho,a}(n) = \binom{2r-2}{\rho-1} n^{-(2r-1)} + O(n^{-2r}). \tag{6.14}$$

Its harmonic pencil has generic rank exactly ρ.

Put $L = r - \rho$ and let

$$V_{L,N}(a) = \det\left(\sum_{n=0}^{N} n^{i+j} R_{r,\rho,a}(n)\right)_{0\le i,j<L}, \qquad V_{0,N} = 1. \tag{6.15}$$

This is rational in N (and rational in a). The degree-ρ homogeneous part of the full r-particle cutoff determinant, with formal harmonic variables X_q corresponding to $H_N^{(q)}(a)$, is exactly

$$\boxed{(-1)^{\rho(\rho-1)/2} V_{L,N}(a) X_\rho^\rho.} \tag{6.16}$$

It is nonzero whenever $N \ge L - 1$, with this cutoff condition vacuous when $L = 0$.

Proof. Positivity follows from (6.9). In (6.10), $h + \rho = 2r - 1$ and $(\rho)_h/h! = \binom{2r-2}{\rho-1}$.

By (6.6), C_ℓ is supported in its final $\rho - \ell$ rows and columns. In its final $(\rho - \ell) \times (\rho - \ell)$ block, every entry below the first possible anti-diagonal in moment order vanishes, and the entries on that anti-diagonal all equal $\binom{h+\ell}{\ell}$. In particular, all matrices lie in the same final $\rho \times \rho$ block. Within that block C_0 has zeros for relative indices $i + j < \rho - 1$ and ones when $i + j = \rho - 1$; all other C_ℓ start strictly after that anti-diagonal. Hence

$$\det\left(\sum_{\ell=0}^{\rho-1} Y_\ell (C_\ell)_{\text{bottom}}\right) = (-1)^{\rho(\rho-1)/2} Y_0^\rho. \tag{6.17}$$

Only the reversing permutation can contribute: its column-index sum already equals the minimum possible sum of nonzero entries in all rows. Thus the pencil has rank ρ.

At $s = \rho$, Y_ℓ is the formal variable $X_{\rho-\ell}$. To obtain degree ρ in the full determinant one must take all these entries from the final block. The complementary determinant is (6.15), which proves (6.16). Its moments have degree at most $2L - 2 < h$, so (6.13) expresses each using finite rational endpoints alone. Positivity of the weight makes this $L \times L$ Gram matrix positive definite once at least L distinct nodes have been included. This proves the last assertion, including the empty case $L = 0$. □

For example, $r = 3, \rho = 2, h = 3$ gives the positive critical weight

$$R(x) = \frac{1}{6}\left\{\frac{1}{(x+a)^2} - \frac{3}{(x+a+1)^2} + \frac{3}{(x+a+2)^2} - \frac{1}{(x+a+3)^2}\right\}.$$

Its two coefficient matrices, indexed by harmonic orders, are

$$C^{(1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad C^{(2)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & -4a-6 \end{pmatrix}.$$

The exact leading harmonic term is $-(\sum_{n=0}^{N} R(n)) X_2^2$, and its coefficient is nonzero for every $N \ge 0$.

6.3 Exact pole orders and the leading Laurent coefficients

For an integer m with $1 \le m \le \rho$, put

$$\eta = \rho - m + 1, \qquad L = r - \eta, \qquad \beta_m = \binom{h+m-1}{m-1}. \tag{6.18}$$

Define the convergent lower-particle normalization

$$G_{L,h}^{(m)}(a) = \det\left(\sum_{n\geq 0} n^{i+j} w_n^{[h]}(m,a)\right)_{0\leq i,j<L}, \qquad G_{0,h}^{(m)} = 1. \tag{6.19}$$

Theorem 6.3 (A staircase of exact spectral poles)**.** *For real $a > 0$, $Z_{r,\rho}(s,a)$ has a pole of exact order $\rho - m + 1$ at every $m = 1, \dots, \rho$. Its leading coefficient is*

$$\boxed{\lim_{s\to m}(s-m)^{\eta} Z_{r,\rho}(s,a) = (-1)^{\eta(\eta-1)/2}\beta_m^{\eta} G_{L,h}^{(m)}(a).} \tag{6.20}$$

The lower normalization is strictly positive. For complex a with $\Re a > 0$, the same identity holds holomorphically, but at special complex parameters the leading coefficient is not asserted nonzero.

Proof. Only $\zeta(s-m+1,a)$ has a pole at $s = m$; its residue is one. Consequently the continued moment matrix has the local form

$$M(s) = \frac{C_{m-1}}{s-m} + A(s)$$

with A holomorphic. The residue matrix is supported in its final $\eta \times \eta$ block. This block has zeros before the anti-diagonal and constant anti-diagonal β_m, so its determinant is $(-1)^{\eta(\eta-1)/2}\beta_m^{\eta}$.

In the full determinant, the coefficient of $(s-m)^{-\eta}$ is this residue determinant times the complementary top-left $L \times L$ minor of $A(m)$. These top-left entries have no pole and coincide with ordinary moments of $w_n^{[h]}(m,a)$. Indeed, their largest degree is $2L-2$ and the weight decays as n^{-m-h}; the required strict inequality is

$$m + h - (2L-2) > 1,$$

which here has left side $\eta + 2 > 1$. The same inequality holds in a small complex neighborhood, so normal convergence identifies the analytic moment entries with their series there. This proves (6.20). At positive real a, the Laplace formula makes the weight strictly positive on infinitely many nodes, so the lower Gram determinant is strictly positive. The pole order is therefore exact. □

6.4 A Gamma product at the highest-order pole

At $m = 1$, put $L = r - \rho$. The normalization in the preceding theorem has the finite product

$$\boxed{G_{L,h}^{(1)}(a) = \prod_{j=0}^{L-1} \frac{(j!)^2 \Gamma(a+j)\Gamma(h-L+1-j)}{\Gamma(h+1-j)\Gamma(a+h-j)}.} \tag{6.21}$$

In particular, the leading Laurent coefficient at $s = 1$ is $(-1)^{\rho(\rho-1)/2}$ times this product. For $\rho = r$ it is the nonzero constant $(-1)^{r(r-1)/2}$.

Here is a derivation that also fixes the normalization. The beta partial-fraction identity gives

$$w_n^{[h]}(1,a) = \frac{\Gamma(n+a)}{\Gamma(n+a+h+1)}.$$

Replace the two monomial bases of the L-particle moment determinant by the monic bases $n^{\underline{i}}$ and $(n+a)_j$. The mixed moment is

$$M_{ij} = \frac{\Gamma(i+1)\Gamma(a+i+j)\Gamma(h-i-j)}{\Gamma(h+1-j)\Gamma(a+h)}, \qquad 0 \leq i,j < L. \tag{6.22}$$

To see this, write $n = i+k$ and sum the resulting Gauss series with parameters $i+1, a+i+j, a+i+h+1$. Its excess is $h - i - j > 0$; in our parameters $h > 2L - 2$. Thus the sum is justified by absolute convergence.

After factoring rows and columns in (6.22), the remaining determinant is

$$\det_{0\le i,j<L} \frac{(a+i)_j}{(1-h+i)_j}.$$

The elementary factorial determinant

$$\det \frac{(x_i)_j}{(x_i+t)_j} = \frac{\prod_{j=0}^{L-1}(t)_j \prod_{i<k}(x_k - x_i)}{\prod_{i=0}^{L-1}(x_i+t)_{L-1}}$$

follows by multiplying each row by $(x_i+t)_{L-1}$ and comparing polynomial coefficients; evaluation at $x_i = -i$ computes the coefficient determinant. Apply it with $x_i = a+i$ and $t = 1-h-a$. Use

$$(1-h-a)_j = (-1)^j \frac{\Gamma(a+h)}{\Gamma(a+h-j)}, \qquad (1-h+i)_{L-1} = (-1)^{L-1} \frac{\Gamma(h-i)}{\Gamma(h-i-L+1)}.$$

All signs cancel, and the remaining factors give (6.21). This is the same classical factorial determinant mechanism as the predecessor's residue product; the new point is its role as the coefficient of a pole of arbitrary order ρ in the shorter-stencil spectral family.

6.5 A complete two-particle formula and its Stieltjes finite parts

For $r = \rho = 2$ one has $h = 1$. Directly from the three moment identities,

$$\boxed{Z_{2,2}(s,a) = 2a^{-s}\zeta(s-1,a) + (1-2a)a^{-s}\zeta(s,a) - \zeta(s,a)^2.} \tag{6.23}$$

Thus, for every integer $m \ge 3$, the ordinary convergent sum

$$\begin{aligned}\sum_{0\le n<k} (k-n)^2\big((n+a)^{-m} - (n+a+1)^{-m}\big)\big((k+a)^{-m} - (k+a+1)^{-m}\big)\\ = 2a^{-m}\zeta(m-1,a) + (1-2a)a^{-m}\zeta(m,a) - \zeta(m,a)^2.\end{aligned} \tag{6.24}$$

For example, $a = 1, m = 3$ gives $2\zeta(2) - \zeta(3) - \zeta(3)^2$.

Use the convention

$$\zeta(1+u,a) = \frac{1}{u} + \sum_{k\ge 0} \frac{(-1)^k \gamma_k(a)}{k!} u^k, \qquad c = \frac{1}{a} - 2.$$

Expanding (6.23) at the double pole gives the full first three Laurent terms

$$\begin{aligned}Z_{2,2}(1+u,a) = &-\frac{1}{u^2} + \frac{c - 2\gamma_0(a)}{u}\\ &+ 2\gamma_1(a) - \gamma_0(a)^2 + c(\gamma_0(a) - \log a + 1) + O(u).\end{aligned} \tag{6.25}$$

At the simple pole $s = 2$,

$$\mathrm{FP}_{s=2}\, Z_{2,2}(s,a) = 2a^{-2}(\gamma_0(a) - \log a) + (1-2a)a^{-2}\zeta(2,a) - \zeta(2,a)^2. \tag{6.26}$$

These may be written entirely in generalized Stieltjes and polygamma coordinates using $\gamma_0(a) = -\psi(a)$ and $\zeta(2,a) = \psi'(a)$. They are spectral Laurent finite parts. No equality with an unsubtracted endpoint sum or a hard-cutoff constant is implicit.

There is a particularly simple check at order zero. Substitution of $\zeta(0,a) = 1/2 - a$ and $\zeta(-1,a) = -(a^2 - a + 1/6)/2$ into (6.23) gives

$$Z_{2,2}(0,a) = \frac{1}{12} \tag{6.27}$$

for its spectral continuation. Yet at $s = 0$ every weight $(n+a)^0 - (n+a+1)^0$ is exactly zero, so the unmodified defining sum at that parameter has ordinary value zero. The discrepancy is an explicit example of noncommutation between continuation and specialization of the original infinite sum; (6.27) does not assign $1/12$ to an ordinary sum of zero terms.

All further Laurent coefficients of (6.8) are obtained by finite determinant arithmetic from the Hurwitz expansions at the relevant shifted orders. In particular, at $s = m$ the coordinates are the generalized Stieltjes constants from the sole resonant Hurwitz factor, together with finitely many ordinary Hurwitz derivatives from the nonresonant factors and elementary endpoint logarithms. This is an exact coefficient-generation procedure, without a claim that the resulting higher constants admit an additional classical reduction.

7 A cutoff polylogarithm identity and its Euler primitive

The first rank-two member of the new hierarchy gives a compact polylogarithmic consequence. Its weight is

$$R(n) = \frac{1}{(n+1)^2} - \frac{1}{(n+2)^2}, \tag{7.1}$$

whereas the predecessor's explicit repeated-pole generator uses the distinct second-difference weight $\frac{1}{2}\{(n+1)^{-2} - 2(n+2)^{-2} + (n+3)^{-2}\}$. Define the finite ordered determinant sum

$$Z_N = \sum_{0 \le n < k \le N} (k-n)^2 R(n) R(k), \qquad Z_0 = 0. \tag{7.2}$$

Our one-direction multiple-polylogarithm convention is

$$\mathrm{Li}_{s_1,\dots,s_j}(z) = \sum_{n_1 > \cdots > n_j \ge 1} \frac{z^{n_1}}{n_1^{s_1} \cdots n_j^{s_j}}, \qquad |z| < 1, \tag{7.3}$$

with positive integer indices. Thus only the largest summation index carries z; in a multiple-argument convention this is $\mathrm{Li}_{s_1,\dots,s_j}(z,1,\dots,1)$. In particular the shuffle identity has normalization $\mathrm{Li}_1(z)^2 = 2\,\mathrm{Li}_{1,1}(z)$.

Theorem 7.1 (Finite cutoff and its complete generator)**.** *For $M = N + 1 \ge 1$,*

$$\boxed{Z_{M-1} = 2\left(1 - \frac{1}{(M+1)^2}\right) H_M - (H_M^{(2)})^2 - \frac{M^2}{(M+1)^2}(H_M^{(2)} + 1).} \tag{7.4}$$

The generating function $G(z) = \sum_{N \ge 0} Z_N z^{N+1}$ satisfies

$$\begin{aligned} z(1-z)G(z) = {} & 2\,\mathrm{Li}_1(z) - \mathrm{Li}_2(z) - z - z\,\mathrm{Li}_4(z) \\ & + 2(1-z)\big(\mathrm{Li}_{1,2}(z) - \mathrm{Li}_{2,1}(z)\big) - (1+z)\,\mathrm{Li}_{2,2}(z). \end{aligned} \tag{7.5}$$

All series are absolutely and locally normally convergent for $|z| < 1$. The quotient in (7.5) is interpreted by its removable value at zero; $G(z) = \frac{5}{48} z^2 + O(z^3)$.

Proof. Use the monic moment basis $1, n+1$, which leaves the determinant unchanged, and put $p = M+1$. Telescoping gives its three moments as

$$\mu_0 = 1 - p^{-2}, \qquad \mu_1 = H_M^{(2)} - Mp^{-2}, \qquad \mu_2 = 2H_M - H_M^{(2)} - 1 + 2p^{-1} - p^{-2}.$$

Expanding $\mu_0\mu_2 - \mu_1^2$ proves (7.4). The basic absolutely convergent summation identities are

$$\sum_{M\geq 1} H_M^{(p)} z^M = \frac{\mathrm{Li}_p(z)}{1-z}, \qquad \sum_{M\geq 1} \frac{z^M H_M^{(p)}}{(M+1)^q} = z^{-1}\,\mathrm{Li}_{q,p}(z),$$
$$\sum_{M\geq 1} (H_M^{(2)})^2 z^M = \frac{2\,\mathrm{Li}_{2,2}(z) + \mathrm{Li}_4(z)}{1-z}, \qquad \sum_{M\geq 1} \frac{z^M}{(M+1)^q} = z^{-1}\,\mathrm{Li}_q(z) - 1.$$

The square identity follows by splitting its two inner indices into the two strict orderings and the diagonal. Substitute these formulas into (7.4), after writing $M^2/(M+1)^2 = 1 - 2/(M+1) + 1/(M+1)^2$. Collecting terms gives (7.5).

Equation (7.4) gives $Z_N = O(\log(N+2))$. Every multiple-polylogarithm series in (7.3) also converges normally on compact subdisks, since its inner sums grow at most polynomially while z^{n_1} decays geometrically. These estimates justify all interchanges. The original generating series proves that the apparent singularity at zero is removable; $Z_0 = 0$ and $Z_1 = R(0)R(1) = 5/48$ give the stated initial terms. □

Corollary 7.2 (An anchored Euler primitive)**.** *For $|z| < 1$, set $I(z) = \int_0^z G(t)\,dt/t$ along any path in the disk. Then*

$$I(z) = \frac{\mathrm{Li}_1(z)^2 + 2\,\mathrm{Li}_{2,1}(z) - \mathrm{Li}_{1,2}(z) + \mathrm{Li}_{2,2}(z) + \mathrm{Li}_2(z) - \mathrm{Li}_1(z)}{z}$$
$$- \mathrm{Li}_5(z) - \mathrm{Li}_{1,4}(z) - 2\,\mathrm{Li}_{3,2}(z) - 2\,\mathrm{Li}_{1,2,2}(z). \tag{7.6}$$

The removable value at zero is $I(0) = 0$, and

$$I(z) = \sum_{N\geq 0} \frac{Z_N}{N+1} z^{N+1} = \frac{5}{96} z^2 + O(z^3).$$

Proof. Let $\theta = z\,d/dz$. From the absolutely convergent series,

$$\theta\,\mathrm{Li}_{p,\boldsymbol{s}} = \mathrm{Li}_{p-1,\boldsymbol{s}} \quad (p > 1), \qquad \theta\,\mathrm{Li}_{1,\boldsymbol{s}} = \frac{z}{1-z}\,\mathrm{Li}_{\boldsymbol{s}}\,.$$

For an empty $\boldsymbol{s}$ the second formula means $\theta\,\mathrm{Li}_1 = z/(1-z)$. These identities also prove $\mathrm{Li}_1^2 = 2\,\mathrm{Li}_{1,1}$ by comparing derivatives and values at zero. Apply them to (7.6): its Euler derivative is the function $G(z)$ specified by (7.5). The convergent expansions at zero give the zero integration constant and the stated removability. Alternatively, coefficientwise division of (7.4) by M gives the same formula by

$$\frac{1}{M(M+1)^2} = \frac{1}{M} - \frac{1}{M+1} - \frac{1}{(M+1)^2}.$$

□

The raw generator has a representation of depth at most two, although its coefficient determinant has harmonic degree two. Its displayed Euler primitive uses depth at most three. These are explicit representations, not claims of minimal depth under functional identities.

7.1 A convergent second Euler endpoint

Corollary 7.3 (An ordinary-zeta endpoint)**.** *The second Euler primitive has the absolutely convergent value*

$$\sum_{N\ge 0}\frac{Z_N}{(N+1)^2}=\int_0^1\frac{-\log z}{z}G(z)\,dz$$
$$=1+3\zeta(2)-2\zeta(3)-\frac{3}{4}\zeta(4)-\zeta(3)^2-\frac{19}{24}\zeta(6). \tag{7.7}$$

The new content here is the determinant endpoint identity; the weight-six multiple-zeta reductions used below are classical.

Proof. The estimate $Z_N=O(\log(N+2))$ proves convergence. Positivity of Z_N also permits Tonelli's theorem in the integral representation. Divide (7.4) by M^2 and collect its rational coefficients to obtain the pointwise identity

$$\frac{Z_{M-1}}{M^2}=-\frac{(H_M^{(2)})^2}{M^2}-\frac{2H_M}{(M+1)^2}+\frac{4H_M}{M(M+1)}-\frac{H_M^{(2)}+1}{(M+1)^2}.$$

Every term is separately summable. The standard nested-series interpretation and elementary telescoping give

$$\sum_{M\ge1}\frac{H_M}{M(M+1)}=\zeta(2),\qquad\sum_{M\ge1}\frac{H_M}{(M+1)^2}=\zeta(2,1)=\zeta(3),\qquad\sum_{M\ge1}\frac{H_M^{(2)}}{(M+1)^2}=\zeta(2,2)=\frac{3}{4}\zeta(4).$$

Splitting coincident and strictly ordered indices gives

$$\sum_{M\ge1}\frac{(H_M^{(2)})^2}{M^2}=2\zeta(2,2,2)+2\zeta(4,2)+\zeta(2,4)+\zeta(6)=\zeta(3)^2+\frac{19}{24}\zeta(6). \tag{7.8}$$

For clarity, the three weight-six inputs are

$$\zeta(2,2,2)=\frac{3}{16}\zeta(6),\qquad\zeta(4,2)=\zeta(3)^2-\frac{4}{3}\zeta(6),\qquad\zeta(2,4)=\frac{25}{12}\zeta(6)-\zeta(3)^2.$$

They follow from short classical calculations. Expanding the cube of $\zeta(2)$ by coincident indices gives the first. Euler's formula $2\zeta(m,1)=m\zeta(m+1)-\sum_{j=1}^{m-2}\zeta(j+1)\zeta(m-j)$, with $m=5$, gives $\zeta(5,1)=\frac{3}{4}\zeta(6)-\frac{1}{2}\zeta(3)^2$; see [15, Eq. (1.3)]. Equating the convergent shuffle and stuffle products

$$\zeta(3)^2=2\zeta(3,3)+6\zeta(4,2)+12\zeta(5,1),\qquad\zeta(3)^2=2\zeta(3,3)+\zeta(6)$$

then gives the second weight-six input. Finally, $\zeta(2,4)+\zeta(4,2)=\zeta(2)\zeta(4)-\zeta(6)$ gives the third. The same Euler formula at $m=2$ supplies $\zeta(2,1)=\zeta(3)$. Substitution into the summable pointwise identity proves (7.7). □

8 Complete reconstruction for smaller tail order ten

The two latest incoming reconstruction reports leave a particularly small but genuine gap. The alternating-tail report proves the order-ten prefix theorem outside $b\equiv962\pmod{2048}$, while the quartic reconstruction report proves it for every $b\ge1920$ [2, 4]. We complete the family. The new

step is a coefficientwise congruence with one additional bit of precision: its only exceptional class is $b \equiv 3010 \pmod{4096}$, which lies entirely inside the range of the second report's real-variable argument. We reproduce that argument at order ten, with explicit constants, to make the resulting nonvanishing proof complete. The small determinant normalization and its Padé origin are credited to the source reports; the modulus-2^{17} congruence and the completed family are the present contribution. No claim of arithmetic independence of numerical special values is involved.

8.1 Statement and exact meaning of reconstruction

Let

$$A_s(n) = \zeta(s, n+1) = \zeta(s) - H_n^{(s)}, \qquad x_n = n + \tfrac{1}{2}, \qquad D_{10,2b}(w) = \sum_{n \geq 0} A_{10}(n) A_{2b}(n) x_n^{-w}.$$

The series initially converges for $\operatorname{Re} w > -2b - 7$. Its centered Euler–Maclaurin continuation is regular at nonpositive even integers. Write $D_r = D_{10,2b}(-2r)$ and use the source's triangular coordinates

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r, \tag{8.1}$$

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j. \tag{8.2}$$

For $j \geq b$, each C_j is a rational combination of $Z_k = \zeta(2k)$, $0 \leq k < b$, with $Z_0 = -1/2$. The words *formal rank* refer to rational coefficient vectors in the formal basis

$$C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1};$$

they do not refer to the linear independence of the values after evaluation.

Put $\mu_j = B_{2j}/(2j)!$ for $j \geq 0$, and $\mu_j = 0$ for $j < 0$. The late coordinate matrix is

$$(M_{5,b})_{tk} = \mu_{t+k-4} \left\{ \frac{(2b+2t-9)!}{(2b-2k)!} + \mathbf{1}_{k<5} \frac{(2t+1)!}{(10-2k)!} \right\}, \qquad 0 \leq t, k < b. \tag{8.3}$$

In particular, $(C_b, \ldots, C_{2b-1})^{\mathsf{T}} = M_{5,b}(Z_0, \ldots, Z_{b-1})^{\mathsf{T}}$. Formula (8.3) also defines the late coordinate row for every integer $t \geq 0$. Its binomial form, useful without factorial conventions, is

$$B_{2t+2k-8} \left\{ \frac{\binom{2t+2}{10-2k}}{2t+2} + \frac{\binom{2b+2t-8}{2b-2k}}{2b+2t-8} \right\},$$

where negative Bernoulli indices give zero and a summand is omitted unless its lower binomial index lies between 2 and its upper index. These are the inherited finite Bernoulli moment formulas [5]. The full first-$2b$ coordinate determinant is $\det M_{5,b}/(4b-1)!!$.

Theorem 8.1 (Complete order-ten optimal prefix)**.** *For every integer $b \geq 6$, $M_{5,b}$ is nonsingular. Consequently, for every even $q > 10$, the q moments*

$$D_{10,q}(0), D_{10,q}(-2), \ldots, D_{10,q}(-2(q-1))$$

have full formal rank q. Every subsequent centered even moment is an explicit rational linear combination of this prefix.

The conclusion now holds for every unequal even pair whose smaller order is at most ten, using the previously established smaller-order families. It does not establish the unrestricted conjecture at smaller orders twelve and above, nor the full gap-six family.

8.2 A finite five-dimensional definition of the numerator

The following normalization is retained from [2]. For $0 \le i, \ell \le 4$, define

$$n_i = b - 5 + i, \quad N_{i\ell} = 2(n_i - \ell) + 1, \quad R_\ell = 2b - 8 + 2\ell,$$

$$F_{i\ell} = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n_i - r)(2n_i + 3 + r),$$

and define Newton coefficients d_{ih} by the finite identity

$$(2n_i + 2 - y)(4n_i + 2 - y)^{\underline{2i}} = \sum_{h=0}^{2i+1} d_{ih} y^{\underline{h}}. \tag{8.4}$$

Equivalently,

$$d_{ih} = \frac{1}{h!} \sum_{r=0}^{h} (-1)^{h-r} \binom{h}{r} (2n_i + 2 - r)(4n_i + 2 - r)^{\underline{2i}}.$$

For independent variables $U_1, \dots, U_9$ put

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)! N_{i\ell}!} \sum_{k=0}^{i} d_{i,2k+1} N_{i\ell}^{\underline{2k+1}} U_{5-i+\ell+k}. \tag{8.5}$$

All factorial ratios with b-dependent arguments have fixed index differences and therefore mean finite rational products in b.

Set

$$\begin{aligned}\Delta_{10}(b) = {} & 4096b^{12} - 110592b^{11} + 1236992b^{10} - 7303680b^9 + 23705088b^8 \\ & - 38140416b^7 + 14925056b^6 + 24490080b^5 - 4469984b^4 \\ & - 15838992b^3 - 2571948b^2 + 4244400b + 1488375,\end{aligned}$$

and define the finite rational expression

$$P_{10}(b; U) = 138240(2b-7)(2b-5)\Delta_{10}(b) \frac{\det \mathcal{A}}{\det F}. \tag{8.6}$$

Exact coefficient arithmetic gives

$$\begin{aligned}\det F = {} & \frac{16384}{4465125}(b-3)(b-2)^2(b-1)(2b-7)(2b-5)^2(2b-3) \\ & \cdot (4b-15)(4b-13)(4b-11)^2(4b-9)^2(4b-7)^2(4b-5)(4b-3)\Delta_{10}(b).\end{aligned} \tag{8.7}$$

The expression P_{10} belongs to $\mathbb{Z}[b, U_1, \dots, U_9]$. It has 439 monomials when grouped by the U exponents, or 10403 monomials after expanding also in b. Its coefficients have degree at most 24 in b. Equations (8.4)–(8.6) specify it completely, without interpolation or a dependence on an unspecified fitted expression. The accompanying exact verifier independently reconstructs (8.7), clears the five row denominators of (8.5), sums the 120 signed permutation products, and compares every coefficient of (8.6) with the inherited integer certificate. Thus the reused certificate is independently replayed as an algebraic identity.

For $b \geq 10$, with $H = (\mu_{t+k-4})_{0 \leq t,k<b}$, define

$$\rho_{5,b} = \frac{\det M_{5,b}}{\dfrac{\prod_{t=0}^{b-1}(2b+2t-9)!}{\prod_{k=0}^{b-1}(2b-2k)!}\det H}.$$

The source reduction is

$$\rho_{5,b} = \frac{\det \mathcal{A}}{\det F}\bigg|_{U_j = 2B_{2b-2j}} = \frac{P_{10}(b; 2B_{2b-2}, \ldots, 2B_{2b-18})}{138240(2b-7)(2b-5)\Delta_{10}(b)}. \tag{8.8}$$

Here H is nonsingular. One direct proof conjugates H by $\operatorname{diag}(1,-1,1,\ldots)$ and partitions after its first three rows and columns. The resulting matrix has blocks

$$\begin{pmatrix} ee^{\mathsf{T}} & B \\ B^{\mathsf{T}} & -P \end{pmatrix}, \qquad e = (0,0,1)^{\mathsf{T}},$$

where P is positive definite by the representation $2\sum_{m\geq 1}(2\pi m)^{-2s}p((2\pi m)^{-2})^2 > 0$ for $s \geq 1$ and a nonzero polynomial p. The first two rows of B have a nonsingular antitriangular minor. Consequently $ee^{\mathsf{T}} + BP^{-1}B^{\mathsf{T}}$ is positive definite.

For completeness, the exact nonzero multiplier behind (8.8) can be obtained without assuming Padé normality. The monic polynomials

$$\pi_n(x) = \sum_{t=0}^{n} \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!}x^t$$

satisfy, for $\mathcal{N}(x^r) = \mu_{r+1}$,

$$\mathcal{N}(x^j\pi_n) = 0 \ (j<n), \qquad \mathcal{N}(x^n\pi_n) = h_n, \quad h_n = \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!}.$$

These finite identities follow by coefficient comparison in the diagonal Padé formula for e^t, as in [4]; they can also be checked directly using the Bernoulli generating function. The same coefficients give

$$\sum_{t=0}^{n}[x^t]\pi_n(x)\,\mu_{t-\ell} = c_n f_\ell(n), \qquad c_n = \frac{(2n+1)(2n+2)!}{(4n+2)!},$$

where $f_\ell(n) = F_{i\ell}$ when $n = n_i$. Apply these monic row combinations to $H + U^{\mathrm{pert}}$, where

$$U^{\mathrm{pert}}_{tk} = \mathbf{1}_{k<5}\mu_{t+k-4}\frac{(2t+1)!(2b-2k)!}{(10-2k)!(2b+2t-9)!}.$$

The last five transformed rows vanish in columns $k \geq 5$. The complementary block is upper triangular with diagonal $h_0, \ldots, h_{b-6}$. Reverse the retained five columns and divide row i by c_{n_i}. The unperturbed entries become F. For the perturbation use the elementary Bernoulli identity

$$\sum_{j=0}^{N}\binom{N}{j}B_j(N-j)^{\underline{h}} = N^{\underline{h}}B_{N-h}(1).$$

Since $b \geq 10$, all the retained indices $N-h$ are at least two; only odd h survive. Restoring the omitted $j=1$ term supplies one further copy of F, explaining the necessary factor $2F$ in (8.5). This proves

$$\det M_{5,b} = (-1)^{5b-15}\frac{\prod_{t=0}^{b-1}(2b+2t-9)!}{\prod_{k=0}^{b-1}(2b-2k)!}\left(\prod_{n=0}^{b-6}h_n\right)\left(\prod_{i=0}^{4}c_{n_i}\right)\det \mathcal{A}. \tag{8.9}$$

The same argument with the perturbation absent gives the identical multiplier for $\det H$ and $\det F$, proving (8.8).

8.3 One additional bit removes the remaining finite obstruction

All congruences in this subsection are in $\mathbb{Z}_{(2)}$, the rationals with odd denominator. We use the classical fact $v_2(B_{2m}) = -1$ for $m \geq 1$.

Lemma 8.2 (A doubled Bernoulli congruence)**.** *For every integer $m \geq 13$,*

$$2B_{2m} \equiv \mathcal{B}_{17}(m) \pmod{131072},$$

where

$$\mathcal{B}_{17}(x) = 1 + 20068x - 26384x^2 + 2816x^3 + 45824x^4 + 4096x^5 - 24576x^6 + 65536x^8. \tag{8.10}$$

Proof. Faulhaber's identity, with $L = 512$, gives

$$\sum_{a=0}^{L-1} a^{2m} = LB_{2m} + \sum_{h=2}^{2m+1} \binom{2m}{h-1} \frac{L^h}{h} B_{2m+1-h}.$$

The $h = 2$ term vanishes. Every term with $h \geq 3$ has valuation at least $9h - v_2(h) - 1 \geq 26$. Divide by 256 and reduce modulo 2^{17}. The even a terms vanish because $2m \geq 26$. For odd a, expand $a^{2m} = (1 + (a^2 - 1))^m$; since $8 \mid a^2 - 1$, terms with binomial degree at least nine have valuation at least $27 - 8 > 17$ after division. Therefore

$$2B_{2m} \equiv \frac{1}{256} \sum_{a=1,3,\ldots,511} \sum_{j=0}^{8} \binom{m}{j} (a^2 - 1)^j \pmod{2^{17}}. \tag{8.11}$$

Expand the finite right side as a polynomial in m. Its rational coefficients all have odd denominators, and their reductions modulo 2^{17}, in increasing powers, are

$$(1, 20068, -26384, 2816, 45824, 4096, -24576, 0, 65536).$$

This is an exact finite coefficient calculation, reproduced by the verifier from the 256 odd support points and the nine displayed binomial terms. It gives (8.10) and proves the lemma. □

Define

$$\begin{aligned} \mathcal{R}(b) = {} & 1536b^8 + 128b^7 - 928b^6 - 816b^5 + 464b^4 \\ & - 1696b^3 - 1930b^2 - 1831b + 694. \end{aligned} \tag{8.12}$$

The new coefficientwise polynomial identity is

$$P_{10}(b; \mathcal{B}_{17}(b-1), \ldots, \mathcal{B}_{17}(b-9)) \equiv 32\mathcal{R}(b) \pmod{131072}. \tag{8.13}$$

This is verified in $\mathbb{Z}[b]$, not only at selected integer arguments. A compact exact procedure is to group the numerator's 10403 terms by their 439 U-monomials, substitute the nine explicit polynomials, multiply modulo 131072, and compare every coefficient with (8.12). The provided script carries out precisely those finite ring operations, after independently rebuilding the numerator from (8.5). Thus (8.13) is a finite algebraic certificate whose entire input is displayed or defined by finite products above.

The linear coefficient of $\mathcal{R}$ is odd, and every coefficient of degree at least two is even. Hence for distinct integers x, y,

$$v_2(\mathcal{R}(x) - \mathcal{R}(y)) = v_2(x - y). \tag{8.14}$$

Indeed, the quotient by $x - y$ is the odd integer -1831 plus an even integer. Direct substitution gives

$$\mathcal{R}(3010) \equiv 0 \pmod{4096}, \qquad \mathcal{R}(962) \equiv 2048 \pmod{4096}.$$

Equation (8.14) proves that 3010 is the unique zero class modulo 4096.

Proposition 8.3 (The new uniform congruence range)**.** *For $b \geq 22$ and $b \not\equiv 3010 \pmod{4096}$, $M_{5,b}$ is nonsingular, and*

$$v_2(\rho_{5,b}) = v_2(b - 3010) - 5. \tag{8.15}$$

Proof. Every $b - j$ in (8.8) is at least 13. Thus (8.2) and (8.13) give

$$\frac{P_{10}(b; 2B_{2b-2}, \ldots, 2B_{2b-18})}{32} \equiv \mathcal{R}(b) \pmod{4096}.$$

Outside the unique zero class, (8.14) gives $v_2(\mathcal{R}(b)) = v_2(b - 3010) < 12$. The numerator therefore has valuation $5 + v_2(b - 3010)$. The denominator in (8.8) has valuation 10: $138240 = 2^{10} \cdot 135$, and all three remaining factors are odd at integer b. This proves the nonvanishing and (8.15). □

8.4 A self-contained sufficient real-variable threshold

We specialize the general highest-degree argument of [4]; this subsection is a credited reproduction with the explicit order-ten constants independently recomputed. Let

$$J(b) = \prod_{i=0}^{4} (2n_i + 1)(2n_i + 2), \qquad E(b) = \prod_{i,\ell=0}^{4} \prod_{r=1}^{\max\{0,2i-4\ell-1\}} (R_\ell + r), \qquad D(b) = 10! J(b) E(b).$$

Here $\deg E = 20$ and $\deg D = 30$. Clearing the fixed factorial and row denominators in (8.5) gives

$$\widetilde{\mathcal{A}} = D\mathcal{A} \in \mathbb{Z}[b, U_1, \ldots, U_9]^{5\times 5}, \qquad \mathscr{P} = \det \widetilde{\mathcal{A}}.$$

Every entry has b-degree at most 46, so $\deg_b \mathscr{P} \leq 230$. If $\|f\|_1$ denotes the sum of absolute values of all coefficients, exact expansion of the 25 entries gives

$$E^* := \max_{i,\ell} \|\widetilde{\mathcal{A}}_{i\ell}\|_1 = 602847821074996042728613866361551228764160.$$

Consequently $\|\mathscr{P}\|_1 \leq K := 120(E^*)^5$. The number E^* is independently recomputed from the entry formulas by the accompanying threshold script; it is not inferred from numerical samples.

Put $F_0 = (2b - 18)!$ and

$$V_j = U_j \frac{F_0}{(2b - 2j)!}, \qquad \kappa = \frac{120(10!)^5}{2!4!6!8!10!} = 14932726824960000000.$$

The part of $\mathscr{P}$ homogeneous of total degree five in U has the exact form

$$\mathscr{P}_{[5]} = -\mathscr{G}(b) \det(V_{i+j+1})_{0\leq i,j\leq 4}, \tag{8.16}$$

where

$$\mathscr{G}(b) = \kappa J(b)^4 E(b)^5 \prod_{i=0}^{4} (4n_i + 2)^{\underline{2i}} \prod_{\ell=0}^{4} \prod_{r=1}^{10+2\ell} (2b - 18 + r).$$

To see the structural factorization, write $T = \mathcal{A} - 2F$ and use $U_j = (2b - 2j)! V_j / F_0$ in (8.5). The identity $N_{i\ell} - (2k + 1) = 2b - 2(5 - i + \ell + k)$ cancels the factorials. The remaining matrix factors as

$$T = \operatorname{diag}\left(\frac{1}{(2n_i + 1)(2n_i + 2)} \right) L \, (V_{5-i+\ell})_{0\leq i,\ell\leq 4} \operatorname{diag}\left(\frac{R_\ell!}{F_0(2\ell + 2)!} \right),$$

where $L_{i,i-k} = d_{i,2k+1}$ is lower triangular. Its diagonal is $d_{i1} = -(i+1)(4n_i+2)^{\underline{2i}}$, which follows directly from the difference of (8.4) at $y = 1$ and $y = 0$. Reverse the five middle rows and multiply all row factors and D^5. This gives exactly (8.16).

At $U_j = 2B_{2b-2j}$, Euler's Bernoulli formula yields

$$|\mathscr{P}_{[5]}| = \frac{\mathscr{G}(b)(4F_0)^5}{(2\pi)^{10b-50}} \det\big(\zeta(2b-18+2i+2j)\big)_{0\le i,j\le 4}. \tag{8.17}$$

The determinant is strictly positive. Indeed, Cauchy–Binet for the discrete moments on m^{-2} gives

$$\det\big(\zeta(2b-18+2i+2j)\big) = \sum_{m_1<\cdots<m_5} (m_1\cdots m_5)^{-2b+18} \prod_{r<s} (m_r^{-2} - m_s^{-2})^2.$$

Keeping $(m_1,\dots,m_5) = (1,2,3,4,5)$ gives the lower bound

$$V\,120^{-2b+18}, \qquad V = \prod_{1\le r<s\le 5} (r^{-2}-s^{-2})^2 = \frac{2401}{36864000000000000}.$$

For $b \ge 20$ every linear factor of $\mathscr{G}$ is at least b, and its degree is 230, so $\mathscr{G}(b) \ge \kappa b^{230}$. For $b \ge 30$,

$$|2B_{2b-2j}| \le T_b := \frac{8(2b)^{16}(2b-18)!}{(2\pi)^{2b-18}}, \qquad 1 \le j \le 9,$$

using $\zeta(2b-2j) < 2$. Also $T_b \ge 1$: since $2b-18 \ge 42$, the elementary comparison $26! > 8^{26}$ and induction suffice. All lower-degree terms of $\mathscr{P}$ have total U-degree at most four. Therefore

$$|\mathscr{P}_{[\le 4]}| \le K b^{230} T_b^4.$$

Divide by (8.17), cancel the powers of 2π before using $2\pi < 8$, and obtain

$$\frac{|\mathscr{P}_{[\le 4]}|}{|\mathscr{P}_{[5]}|} \le R(b) := C\frac{b^{64}960^{2b}}{(2b-18)!}, \qquad C = \frac{K8^{26}2^{64}}{\kappa 4^5 V 120^{18}}. \tag{8.18}$$

Every constant is explicit and rational. Exact integer comparison after clearing denominators gives

$$R(1920) < 2^{-546} < 1.$$

The verifier checks this stronger margin. Moreover, for $b \ge 1920$,

$$\frac{R(b+1)}{R(b)} = \left(1+\frac{1}{b}\right)^{64} \frac{960^2}{(2b-16)(2b-17)} < \frac{3}{4}.$$

For a simple uniform justification, $(1+1/b)^{64} < 3$, both denominator factors exceed b, and $b \ge 1920 = 2\cdot 960$. Thus the highest-degree term cannot be cancelled for any $b \ge 1920$. Since $D(b) > 0$ and the multiplier (8.9) is nonzero, we have proved

$$\det M_{5,b} \ne 0 \qquad (b \ge 1920). \tag{8.19}$$

8.5 Completion, and an independent audit of the old exception

For $22 \le b < 1920$, one cannot have $b \equiv 3010 \pmod{4096}$. Proposition 8.3 therefore proves nonvanishing throughout this interval. Equation (8.19) supplies all $b \ge 1920$. The remaining $6 \le b \le 21$ are checked directly from the rational source matrix (8.3); the machine-readable record contains every nonzero determinant. Most of these base cases are already contained in [2]; they are recomputed here. For example,

$$\det M_{5,6} = -\frac{7097067}{2867200}, \qquad \det M_{5,8} = \frac{537653622905129}{68669669376000}.$$

This proves Theorem 8.1.

The old exceptional index $b = 962$ also has a short independent exact audit. Its numerator satisfies

$$P_{10}(962; 2B_{1922}, 2B_{1920}, \dots, 2B_{1906}) \equiv 2386755584 \pmod{2^{32}}. \tag{8.20}$$

The residue has valuation 16, giving $v_2(\rho_{5,962}) = 6$. Two independent calculations of the nine Bernoulli residues agree: the recurrence

$$(2m+1)(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1} \binom{2m+1}{2j} (2B_{2j}),$$

with odd division modulo 2^{32}, and the Faulhaber power sum over the 32768 odd integers below 65536, performed modulo 2^{47} before dividing by 2^{15}. Finally, evaluating the rational five-dimensional determinant (8.5) directly, with these integer residues as inputs, gives the same residue (8.20) as the sparse numerator. This audit is not needed once the uniform congruence is proved, but it verifies the exact location where the two preceding reports failed to overlap.

8.6 An ordinary convergent family of polygamma identities

For every $b \ge 6$ and $0 \le k < b$, let

$$Q_{b,k}(x) = \sum_{t=0}^{b-1} (M_{5,b}^{-1})_{kt} \big((x + \tfrac{1}{2})^{2b+2t+1} - (x - \tfrac{1}{2})^{2b+2t+1} \big). \tag{8.21}$$

This form is shorter than an expansion of the even polynomial. The identity

$$(x + \tfrac{1}{2})^{2j+1} - (x - \tfrac{1}{2})^{2j+1} = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} x^{2r}$$

shows that (8.21) is precisely the inverse coordinate combination dictated by (8.1).

Define

$$\eta_{s,0} = \frac{1}{s-1}, \qquad \eta_{s,j} = \frac{(s)_{2j-1} B_{2j}(1/2)}{(2j)!} \quad (j \ge 1), \qquad c_{b,\ell} = \sum_{j=0}^{\ell} \eta_{10,j} \eta_{2b,\ell-j}.$$

Let $S_{b,k}(x)$ be the polynomial consisting of the nonnegative powers in the formal expansion

$$Q_{b,k}(x) x^{-2b-8} \sum_{\ell \ge 0} c_{b,\ell} x^{-2\ell}.$$

It is an even polynomial of degree at most $2b - 10$, with rational coefficients determined by finitely many terms.

Corollary 8.4 (All order-ten reconstruction identities)**.** *For every integer $b \geq 6$ and $0 \leq k < b$,*

$$\boxed{\sum_{n=0}^{\infty} \left\{ Q_{b,k}(x_n) \frac{\psi^{(9)}(n+1)\psi^{(2b-1)}(n+1)}{9!\,(2b-1)!} - S_{b,k}(x_n) \right\} = \zeta(2k).} \tag{8.22}$$

The series is ordinary and absolutely convergent.

Proof. For even s, $A_s(n) = \psi^{(s-1)}(n+1)/(s-1)!$. By (8.1), (8.21), and the exact matrix inverse, the continued sum of $Q_{b,k}(x_n)A_{10}(n)A_{2b}(n)$ is $\zeta(2k)$. The centered Euler–Maclaurin expansion of the product is exactly the formal product used to define $S_{b,k}$. Removing all nonnegative powers leaves $O(x_n^{-2})$. Every removed power is even and has centered continued sum $\zeta(-2r, 1/2) = 0$, $r \geq 0$. Therefore the continued sum agrees with the displayed absolutely convergent series. □

Here is a completely explicit member, included to make the reconstruction mechanism concrete. Let

$$\begin{aligned} P(x) = {} & 58535757x^{23} + 145331415x^{21} - 98765059x^{19} \\ & + 92514527x^{17} - 73310041x^{15} + 28468113x^{13}. \end{aligned}$$

Then

$$\begin{aligned} \sum_{n=0}^{\infty} \Bigg\{ & [P(n+1) - P(n)][\zeta(10) - H_n^{(10)}][\zeta(12) - H_n^{(12)}] \\ & - \frac{149591379(n+\frac{1}{2})^2 + 1835020425}{11} \Bigg\} = -\frac{574862427}{8}. \end{aligned} \tag{8.23}$$

It follows from the $k = 0$ row of $M_{5,6}^{-1}$:

$$\frac{4}{574862427}(28468113, -73310041, 92514527, -98765059, 145331415, 58535757).$$

The numerical pair $(10, 12)$ was already within the source's finite range; the explicit identity is an illustration of the now-uniform construction, not the new nonvanishing case. All coefficients, counterterms, and the exact inverse-row check are included in the identity artifact.

Further questions specific to this completion. The unrestricted rank problem remains open in the inspected corpus for smaller order at least twelve. A natural next question is whether one can prove a general overlap principle between the finite 2-adic exceptional classes and the constructive real-variable threshold, without computing an increasingly large numerator. Another is whether a determinant representation makes the lower Bernoulli-degree terms sign-compatible, giving nonvanishing directly and avoiding a domination threshold. The gap-six family, where the generic nearby-Bernoulli contraction falls outside its stable range, still calls for a separate argument. Finally, one may seek compact closed forms for the inverse rows in (8.21); the existence and rational computability proved here do not by themselves make those coefficients small.

9 A cyclotomic test case already determined by the manuscript

The incoming alternating-tail report asks, as the first test in its complementary-sector Question 7, for $\operatorname{Im} Z(2,1;i,-i)$, where

$$Z(2,1;i,-i)=\sum_{k>m\geq 1}\frac{i^k(-i)^m}{k^2m}.$$

The full two-resolvent complementary-sector generator remains open. However, this particular coefficient is already an exact consequence of two results in the pinned unified manuscript. Recording that implication is an integration correction, not a newly discovered period evaluation.

Proposition 9.1 (Explicit complementary coefficient)**.** *Put* $\lambda_3=\operatorname{Im}\operatorname{Li}_3((1+i)/2)$. *Then*

$$\boxed{\operatorname{Im} Z(2,1;i,-i)=\lambda_3-\frac{\pi\log^2 2}{32}-\frac{\pi^3}{128}.} \tag{9.1}$$

Equivalently,

$$\int_0^1\frac{\log t\log(1-t)}{1+t^2}\,\mathrm{d}t=\lambda_3-\frac{\pi\log^2 2}{32}-\frac{\pi^3}{128}. \tag{9.2}$$

Proof. Use the one-direction convention $\operatorname{Li}_{a,b}(z)=\sum_{k>m\geq 1}z^k/(k^am^b)$. The inverse-color theorem in `chapters/04-depth.tex`, equation `inverse:eq:mixedb1`, gives

$$Z(2,1;z,z^{-1})=2\operatorname{Li}_3(z)+\operatorname{Li}_{2,1}(z)+\operatorname{Li}_{1,2}(z)-\zeta(2)\operatorname{Li}_1(z).$$

The iterated-integral shuffle identity is $\operatorname{Li}_1(z)\operatorname{Li}_2(z)=\operatorname{Li}_{1,2}(z)+2\operatorname{Li}_{2,1}(z)$. Hence

$$Z(2,1;z,z^{-1})=2\operatorname{Li}_3(z)+\operatorname{Li}_1(z)\operatorname{Li}_2(z)-\operatorname{Li}_{2,1}(z)-\zeta(2)\operatorname{Li}_1(z). \tag{9.3}$$

At $z=i$, put $L=\log 2$ and $G=\beta(2)$. The standard values are

$$\operatorname{Li}_1(i)=-L/2+i\pi/4,\quad \operatorname{Li}_2(i)=-\pi^2/48+iG,\quad \operatorname{Im}\operatorname{Li}_3(i)=\pi^3/32.$$

The proved height-one specialization in `chapters/04-shuffle-parity.tex` is

$$\operatorname{Im}\operatorname{Li}_{2,1}(i)=-GL/2+\pi L^2/32-\lambda_3+3\pi^3/128.$$

Substituting these identities into (9.3) cancels the Catalan terms and yields (9.1). Both inherited reductions are proved in the source, not inferred from a numerical fit [1].

For the integral form, use the same manuscript's inverse-color integral representation, or insert the elementary Mellin formula for k^{-2} directly in the absolutely convergent double sum. It gives

$$Z(2,1;i,-i)=i\int_0^1\frac{\log t\log(1-t)}{1-it}\,\mathrm{d}t.$$

Taking the imaginary part proves (9.2). The endpoint logarithms are integrable and $1-it$ does not vanish on the integration interval, so no limiting branch choice is hidden here. □

The numerical value is

$$0.280672506182661124396068053604527275428 4\ldots.$$

This is a finite reduction into the explicitly stated classical trilogarithm collection. It does not reduce λ_3 further and does not prove its independence from any smaller set of constants. Nor does it settle the full complementary generator or the distinct Gaussian S_6 and revised S_8 conjectures.

The proposed status update is therefore precise: retain Question 7 for the full complementary family, mark its displayed first coefficient as already determined by the manuscript, and include (9.1) as the missing cross-reference. The source proofs of inverse-color transport and height-one reduction deserve the credit.

10 Further research questions with concrete mathematical targets

The following are proposals for further work, not consequences claimed in the preceding theorems. They prioritize exact structural statements and finite evaluations. A question asking for a classification allows a counterexample with a proved exceptional set as a useful answer.

10.1 Reconstruction beyond order ten

Question 1 (All unequal even orders)**.** For every $1 \le a < b$, is the first-$2b$ centered even-moment prefix of the $(2a, 2b)$ tail product nonsingular in its formal source coordinates? The present article completes $a = 5$; the project now has all $a \le 5$. A decisive next result would be a structural noncancellation theorem for the fixed-size Bernoulli determinant, or a proved description of any exceptional pairs.

Question 2 (A uniform overlap principle)**.** Can one relate the precision of a doubled-Bernoulli congruence to the constructive real-variable threshold for arbitrary fixed a? At $a = 5$ the sole exceptional class has least positive representative 3010, exceeding the threshold 1920. Higher precision alone does not prove nonvanishing: a compatible 2-adic root may persist. The target is a theorem ensuring coverage by two complementary arguments, not a table of increasingly long nonzero prefixes.

Question 3 (The full gap-six family)**.** Give a bordered-Hankel reduction that retains the moving B_0 and B_1 endpoint terms for $b = a + 3$. The stable fixed-order contraction requires $b \ge 2a$ and therefore does not cover this family uniformly. A formula with all boundary terms retained could settle the global gap-six conjecture without extrapolating the finitely checked pairs.

Question 4 (Compact inverse rows)**.** Find a product, determinant of uniformly smaller size, or short recurrence for the polynomials $Q_{b,k}$ of Corollary 8.4. Their exact rational computability is proved, but the displayed $(10, 12)$ example shows that naive expanded coefficients grow rapidly. A compact expression should also identify the subtraction polynomial directly and preserve ordinary convergence.

10.2 Beyond the bilinear Gamma normalization

Question 5 (Every bilinear contact hyperplane)**.** For the two independent orders in Theorem 2.1, compute the constant terms of C_N and $(-1)^N D_N$ on the further resonant hyperplanes $p + q = -2m$. The first symmetric contact is $1/2$ at $p + q = 0$. Seek closed bivariate contact polynomials, every independent p, q jet, and the exact change under translating the cutoff. The diagonal answer in Section 4 is a mandatory check; it does not determine all transverse derivatives.

Question 6 (Independent outer weights and ordinary Hurwitz tails)**.** Construct a finite-cutoff identity for $\zeta(p, n + a)\zeta(q, n + b)(n + c)^{-w}$ with independent p, q, w, a, b, c, retaining an explicit remainder or a precisely defined finite collection of additional functions. Determine its moving polar hyperplanes and residues before differentiating on them. The Gamma-weighted alternating identities do not settle this ordinary Hurwitz problem; a valid result must state the size and analytic meaning of any newly required coordinate collection.

Question 7 (More colors and higher products)**.** For tails with a primitive dth-root twist, can a product of Gamma quotients convert the d-step recurrence into a finite bilinear endpoint law? For three tail factors, identify the smallest adjacent product ansatz whose linear and quadratic residual terms can all cancel. A useful target is a divisor-orbit criterion extending Theorem 3.1, together with one ordinary convergent three-factor evaluation that genuinely lies outside its bilinear case.

Question 8 (The remaining logarithmic Gamma generator)**.** The logarithmic square sum in Corollary 4.6 is explicit, but the earlier twisted logarithmic-Gamma correction generator has an independent outer resolvent. Determine whether it belongs to a finite stated collection of classical or multiple-Gamma functions, or prove a functional obstruction in a declared difference-equation class. A failure of numerical reduction is not such an obstruction.

10.3 Harmonic pencils, spectral coefficients, and polylogarithms

Question 9 (Complex weights and sharp cutoff exceptions)**.** Does generic degree–rank equality extend to every complex rational weight? If not, classify the smallest counterexamples using the kernel-compression identity (5.4). For real weights, determine whether the bound $r(r-\rho)$ on late deficient cutoffs is sharp in the rational-moment class, and how it improves for special polynomial kernel spaces. The real proof uses definite increments; it cannot be transferred to a complex bilinear Gram form by wording alone.

Question 10 (Higher-rank pencils and their realizations)**.** Classify rational residue data whose entire harmonic pencil has generic rank two, then rank ρ. The family in Theorem 6.2 realizes every rank, but does not classify all pencils: the maximal-rank kernel need not be a common kernel. Seek exact finite residue conditions and identify which such pencils come from positive rational weights. This would complement the existing rank-one classification.

Question 11 (All Laurent coefficients in compact form)**.** For the hierarchy $Z_{r,\rho}(s,a)$, find short formulas or recurrences for the lower normalizations $G_{L,h}^{(m)}(a)$ at $m \geq 2$, and for the complete principal and finite parts at every pole. The leading coefficient at $m=1$ is a Gamma product. At higher orders, confluent deformations of the mixed factorial moments may reduce determinant growth. Any claimed formula should reproduce both finite parts in (6.25)–(6.26) and distinguish them from hard-cutoff constants.

Question 12 (Complementary cyclotomic generators and exact depth)**.** Construct the full complementary two-resolvent generator requested in the alternating report, retaining the evaluated coefficient (9.1) as a boundary check. In parallel, determine minimal depth for concrete harmonic-determinant generating functions inside a specified shuffle, differential, or symbol calculus. Formal harmonic degree predicts a useful representation bound; it does not by itself prove minimal polylogarithmic depth. The Gaussian S_6 and revised S_8 assertions remain separate conjectures requiring exact reductions or counterexamples.

10.4 A practical order of attack

The next finite target is smaller order twelve: derive its exact fixed-size numerator, determine a sufficiently precise Bernoulli congruence, and compare the exceptional classes with the real-variable threshold. The next analytic target is the first off-diagonal contact hyperplane $p+q=-2$, where the new bilinear endpoint contains enough independent parameters to reveal which diagonal parity cancellations survive. The next determinant target is a rank-two residue classification, tested against both the positive shorter-stencil family and a pencil without a common kernel.

For formal verification, the finite bilinear recurrence, rational root-orbit criterion, and symmetric-pencil compression lemma are small natural starting points. The order-ten congruence can then be checked in a polynomial ring modulo 2^{17} from the finite Newton matrix. Such formalization would strengthen the trust chain without altering the claimed mathematics or replacing a proof by decimal agreement.

11 Verification, source corrections, and integration

11.1 Proofs and exact computational certificates

The bilinear, rational-certificate, cutoff-contact, harmonic-degree, and spectral-hierarchy theorems have proofs in the article. Their finite formulas also admit independent symbolic and numerical replays. The order-ten completion additionally uses an exact finite computer-assisted certificate: a polynomial defined by a 5×5 Newton/factorial determinant is reconstructed coefficientwise, then substituted into an explicit congruence in $\mathbb{Z}[b]/(2^{17})$. All of the inputs, finite algorithms, and integer comparisons are included. The proof does not depend on floating-point nonzero tests.

The complete polynomial reconstruction sums all 120 signed permutation products and matches 10 403 expanded coefficients, stored in 439 groups. The old exceptional integer $b = 962$ is checked separately by two Bernoulli computations and by direct evaluation of the 5×5 determinant. The sufficient threshold $b = 1920$ is regenerated from the source entry formulas and compared using exact integers. Finite source matrices handle $6 \le b \le 21$.

11.2 Reproducible diagnostics

The included programs use Python, SymPy, and mpmath; the observed versions were Python 3.12, SymPy 1.14.0, and mpmath 1.3.0. They need no network access. The package README supplies commands and file locations. The separate result records preserve the precision and test method rather than presenting all checks as equivalent evidence.

Component	Exact checks or independent route	Role
Bilinear tails	Five symbolic cancellations; 28 numerical checks, including independent Laplace integrals for Stieltjes jets.	Recurrence-sign, normalization, domain, and example diagnostics.
Cutoff jets and cyclotomic audit	74 exact Euler-polynomial and translated-coefficient checks through order twelve; 12 numerical checks, including independent quadrature.	Contact signs, sharp threshold, differentiated finite sums, and the inherited trilogarithm coefficient.
Degree and spectral hierarchy	45 exact checks, including all ranks for $r = 2, \dots, 6$; 19 numerical checks through independent Mellin moments and Laurent limits.	Non-common-kernel example, signed cutoff exception, pole normalization, Gamma products, and Stieltjes finite parts.
Polylogarithm generator	122 exact cutoff and Taylor-coefficient checks; independent endpoint evaluation from direct moments and harmonic tail expansions.	Generator, anchored primitive, and ordinary-zeta endpoint; 65-digit numerical diagnostics.
Order-ten completion	Full coefficient reconstruction, uniform congruence, exact base matrices and threshold; two independent Bernoulli residue calculations.	Exact finite certificates supporting the uniform nonvanishing theorem.
Explicit $(10, 12)$ identity	Exact inverse-row and counterterm arithmetic; independent numerical cutoff/tail diagnostics.	An ordinary convergent illustration of the uniform family.

The 45-digit bilinear checks have residuals below $3.1 \cdot 10^{-45}$. Independent Mellin comparisons for the determinant formulas agree at roughly 10^{-79} relative error at 80-digit precision; the finite-step

Laurent extrapolations have residuals of order 10^{-24}. The polylogarithm endpoint checks have errors below $2 \cdot 10^{-41}$ and $4 \cdot 10^{-48}$ with two truncation orders. The displayed $(10, 12)$ identity has numerical residuals below $4 \cdot 10^{-68}$. These are numerical diagnostics, not certified interval bounds. They neither prove global identities nor establish independence of constants. The written arguments and exact algebraic certificates have that separate role.

Independent mathematical audits checked the bilinear endpoint normalization, the rational factor $cS(0)$, translated cutoff jets, the pencil compression argument, the finite-difference residue blocks, and the Gamma-product normalization. No unresolved discrepancy was found in those checks. This records the verification performed; it does not describe the article as externally refereed or formally verified in a proof assistant.

11.3 Corrections and status updates proposed for ProveIt

The concrete correction is to the first coefficient in the incoming complementary-sector question. Add the formula in Proposition 9.1 and cross-reference the existing inverse-color and height-one manuscript reductions. Retain the full generator question. The correction concerns an overlooked implication, not an incorrect proved identity.

The order-ten progression should now be marked proved by Theorem 8.1; its prior congruence remains valid on its original range. The new result does not invalidate that earlier proof. The degree–rank question should be split into the proved real-rational generic theorem, the explicit finite-cutoff exception test, and the remaining complex/depth questions. The ordinary-square continuation question is resolved with its cutoff coordinate included in the statement.

Two presentation precautions are mathematically necessary. First, "odd negative orders have no contact" is true for $X = N + a + 1$; it is false for an arbitrary translated cutoff. Second, all-coordinate formal rank is not arithmetic independence of zeta values. These qualifications should travel with the imported statements.

No new error in the already proved core manuscript is asserted by this audit, and this work is not an exhaustive audit of the whole repository. The Gaussian S_6 and revised S_8 statements are not proved here. The unrestricted even-order prefix theorem, complete gap-six family, ordinary independent-order outer-weight problem, and full complementary generator remain explicitly open.

11.4 Integration boundary

The package supplies a modular `article.tex`, section sources, a flattened standalone TeX file, the rendered PDF, exact verification programs and records, and source/status notes. Section labels have disjoint prefixes so individual results can be imported. The supplied source pin and question map identify what was inspected. The work does not modify the repository, supersede all earlier reports, or bundle their generated build files.

References

[1] V. Reshetnikov, ProveIt repository, unified polylogarithm manuscript. Inspected revision `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`, 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt. Relevant manuscript files are `Analysis/Polylogarithms/docs/manuscript/chapters/04-depth.tex` and `chapters/04-shuffle-parity.tex` relative to that manuscript.

[2] ProveIt research corpus, *Alternating Tails and Higher Spectral Defects*, incoming package `ProveIt_Alternating_Tails_Spectral_Defects_2026-10-11.zip`, at the revision in [1].

Sections on polynomial adjacent-tail certificates and order-ten congruences; further-research Questions 1–7.

[3] ProveIt research corpus, *Finite Difference Determinants*, incoming package `ProveIt_Finite_Difference_Determinants_2026-10-11.zip`, at the revision in [1]. Relevant sections: `sections/02-spectral.tex`, `sections/06-classification.tex`, `sections/09-questions.tex`, and `sections/10-residue-product.tex`.

[4] ProveIt research corpus, *Harmonic-Tail Reconstruction, Quartic Trigamma Moments, and Spectral Stieltjes Calculus*, incoming package `ProveIt_Quartic_Spectral_Reconstruction_2026-10-11.zip`, at the revision in [1]. Sections 2–3 supply the inherited order-eight completion and the general constructive eventual-rank argument.

[5] ProveIt research corpus, *Bernoulli–Tornheim–Digamma continuation*, "A fixed-size determinant for every pair of even orders," `sections/03b_general_hankel.tex`, in `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`, at the revision in [1].

[6] ProveIt research corpus, incoming package `ProveIt_Higher_Quadratic_Jets_2026-10-11.zip`, at the revision in [1]. Inspected as prior context; its higher-order jet constructions are not claimed as new in the present article.

[7] ProveIt research corpus, incoming package `ProveIt_Higher_Spectral_Defects_2026-10-11.zip`, at the revision in [1]. Inspected as prior context; the present shorter-stencil family is distinguished from its general spectral-defect questions.

[8] S. Hu and M.-S. Kim, *On the Stieltjes constants and gamma functions with respect to alternating Hurwitz zeta functions*, J. Math. Anal. Appl. **509** (2022), no. 1, article 125930. https://arxiv.org/abs/2106.14674.

[9] W. Wang, S. Hu, and M.-S. Kim, *On gamma functions with respect to the alternating Hurwitz zeta functions*, arXiv:2405.02854v4 (2024). https://arxiv.org/abs/2405.02854v4.

[10] S. Hu and M.-S. Kim, *Asymptotic expansions for the alternating Hurwitz zeta function and its derivatives*, arXiv:2103.15528v6 (2023). https://arxiv.org/abs/2103.15528. The underlying uniform expansions, Euler specializations, and spectral derivative calculus are classical input here.

[11] M. E. Hoffman, *Sums of products of Riemann zeta tails*, Mediterr. J. Math. **13** (2016), 2771–2781. https://doi.org/10.1007/s00009-015-0653-9; https://arxiv.org/abs/1602.03399.

[12] S. A. Abramov, *The rational component of the solution of a first-order linear recurrence relation with rational right-hand side*, U.S.S.R. Comput. Math. Math. Phys. **15** (1975), no. 4, 216–221. https://doi.org/10.1016/0041-5553(75)90181-0.

[13] NIST Digital Library of Mathematical Functions, Section 25.11, Hurwitz zeta function. https://dlmf.nist.gov/25.11.

[14] NIST Digital Library of Mathematical Functions, Sections 5.5 and 5.11, Gamma functional relations and asymptotic expansions. https://dlmf.nist.gov/5.5; https://dlmf.nist.gov/5.11.

[15] Jonathan M. Borwein and David M. Bradley, *On Two Fundamental Identities for Euler Sums*, preprint, January 2005, especially Eq. (1.3). https://carmamaths.org/resources/jon/Preprints/Papers/ArXiv/Zeta21/z21.pdf.